

Exercises

Level - 1

(Problems Based on Fundamentals)

First Principle of Differentiation

1. Find the derivative of $y = e^x$, using first principle.
2. Find the derivative of $y = f(x) = \log_e x$, using first principle.
3. Find the derivative of $y = f(x) = \log_a x$, using first principle.
4. Find the derivative of $y = f(x) = \sin x$, using first principle.
5. Find the derivative of $y = f(x) = \tan^{-1} x$, using first principle of differentiation.
6. Find the derivative of $y = f(x) = e^{\sin x}$, using first principle.
7. Find the derivative of $y = f(x) = \sin(x^2)$, using first principle.

ABC of Differentiation

8. If $y = \log_x x + 10$, find $\frac{dy}{dx}$
9. If $y = 5^{\log_3 x} - x^{\log_y 5}$, find $\frac{dy}{dx}$
10. If $y = \frac{1}{x} + \frac{2}{x^2} + \frac{3}{x^3}$, find $\frac{dy}{dx}$
11. If $y = \sqrt{x} + x\sqrt{x} + x^2\sqrt{x} + x^3\sqrt{x}$, find $\frac{dy}{dx}$
12. If $y = \left(1 + \tan\left(\frac{\pi}{8} - x\right)\right)\left(1 - \tan\left(x + \frac{\pi}{8}\right)\right)$, find $\frac{dy}{dx}$

13. If $y = \frac{\sec x + \tan x - 1}{\sec x - \tan x + 1}$, find $\frac{dy}{dx}$ at $x = 0$
14. If $\frac{x^4 + x^2 + 1}{x^2 - x + 1}$ such that $\frac{dy}{dx} = ax + b$, find the value of $a + b + 10$
15. If $y = \left(\sin \frac{x}{2} + \cos \frac{x}{2}\right)^2$, find $\frac{dy}{dx}$ at $x = \frac{\pi}{6}$
16. If $y = \left(\sqrt{\frac{x}{a}} + \sqrt{\frac{a}{x}}\right)$, prove that $2xy \frac{dy}{dx} = \frac{x}{a} - \frac{a}{x}$

Differentiation of Composite Functions

17. If $y = \log(\sin(3x + 5))$, find $\frac{dy}{dx}$
18. If $y = \log(x + \sqrt{x^2 + 1})$, find $\frac{dy}{dx}$
19. If $y = \log(\sqrt{x-1} - \sqrt{x+1})$, find $\frac{dy}{dx}$
20. If $y = (\sin x)^x$, find $y = (\sin x)^x$
21. If $y = \log(\sin x + \cos x)$, find $\frac{dy}{dx}$
22. If $f(x) = \frac{1}{x-1}$, find $\frac{d(f(f(x)))}{dx}$
23. If $f(x) = \frac{x-1}{x+1}$, find $\frac{d(f(f(x)))}{dx}$
24. Let f be a function for which $f'(x) = \frac{1}{x^2 + 1}$
If $g(x) = f(3x - 1)$, find $g'(x)$.
25. Let f be a function for which $f'(x) = x^2 + 1$
If $y = f(\sin(x^3))$, find $\frac{dy}{dx}$.
26. If $f(x) = |x-1| + |x-3|$, find $f'(2)$
27. If $f(x) = |x^2 - 1| + |x^2 - 4|$, find the value of $f'\left(\frac{3}{2}\right)$.
28. If $f(x^2) = x^4 + x^3 + 1$, find $f'(x^4)$
29. If g is the inverse of f and $f'(x) = \cos 2x$, find $g'(x)$.
30. If $f(x) = x + \tan x$ and f is the inverse of g , find $g'(x)$.
31. If $y = \frac{x^4 - x^2 + 1}{x^2 + \sqrt{3}x + 1}$ such that $\frac{dy}{dx} = px + q$ then prove that $p - q = \tan\left(\frac{5\pi}{12}\right)$
32. If $y = \sqrt{x-1} + \sqrt{x+1}$, prove that $\sqrt{x^2 - 1} \frac{dy}{dx} = \frac{1}{2}y$.
33. If $y = \frac{x}{x+2}$, prove that $x \frac{dy}{dx} = (1-y)y$.

34. If $y = \sqrt{x} + \frac{1}{\sqrt{x}}$, prove that $2x \frac{dy}{dx} = \sqrt{x} - \frac{1}{\sqrt{x}}$.

35. If $y = e^x \cos x$, prove that $\frac{dy}{dx} = \sqrt{2} e^x \cos\left(x + \frac{\pi}{4}\right)$.

36. If $y = \sqrt{x^2 + a^2}$, prove that $y \frac{dy}{dx} - x = 0$.

37. If $y = e^x + e^{-x}$, prove that $\frac{dy}{dx} = \sqrt{y^2 - 4}$.

38. If $xy = 4$, prove that $x \left(\frac{dy}{dx} + y^2\right) = 3y$.

39. Prove that

$$\frac{d}{dx} \left\{ \frac{x}{2} \sqrt{a^2 - x^2} + \frac{a^2}{2} \sin^{-1} \left(\frac{x}{a} \right) \right\} = \sqrt{a^2 - x^2}$$

Differentiation by Inverse Trigonometric Function

40. Differentiate the following functions w.r.t. x :

(i) $\tan^{-1}(\sqrt{1+x^2} + x)$

(ii) $\tan^{-1}(\sqrt{1+x^2} - x)$

(iii) $\tan^{-1} \left\{ \frac{\sqrt{1+x^2} - 1}{x} \right\} \quad x \neq 0$

(iv) $\tan^{-1} \left\{ \frac{\sqrt{1+x^2} + 1}{x} \right\} \quad x \neq 0$

(v) $\cot^{-1}(\sqrt{1+x^2} + x)$

(vi) $\tan^{-1} \left\{ \frac{\sqrt{1+\sin x} + \sqrt{1-\sin x}}{\sqrt{1+\sin x} - \sqrt{1-\sin x}} \right\}, 0 < x < \pi$

41. Differentiate the following functions w.r.t. x :

(i) $\tan^{-1} \left(\frac{a+x}{1-ax} \right)$

(ii) $\tan^{-1} \left\{ \frac{a \cos x - b \sin x}{a \cos x + b \sin x} \right\}, -\frac{\pi}{2} < x < \frac{\pi}{2}$

(iii) $\tan^{-1} \left(\frac{3a^2x - x^3}{a^3 - 3ax^2} \right), -\frac{1}{\sqrt{3}} < \frac{x}{a} < \frac{1}{\sqrt{3}}$

(iv) $\tan^{-1} \left(\sqrt{\frac{a-x}{a+x}} \right), -a < x < a$

(v) $\tan^{-1} \left(\frac{\sin x}{1 + \cos x} \right), -\pi < x < \pi$

(vi) $\tan^{-1} \left(\frac{a + b \tan x}{b - a \tan x} \right)$

(vii) $\tan^{-1} \left(\frac{a + bx}{b - ax} \right)$

(viii) $\tan^{-1} \left(\frac{x}{1 + 6x^2} \right)$

(ix) $\sin^{-1} \left(\frac{\sin x + \cos x}{\sqrt{2}} \right), -\frac{3\pi}{4} < x < \frac{3\pi}{4}$

- (x) $\sin^{-1}\left(\frac{1-x^2}{1+x^2}\right) + \sec^{-1}\left(\frac{1+x^2}{1-x^2}\right)$, $x \in R$
- (xi) $\sin^{-1}\left(\frac{x-1}{x+1}\right) + \cos^{-1}\left(\frac{x-1}{x+1}\right)$
- (xii) $\sin^{-1}\left(\frac{3\sin x + 4\cos x}{5}\right)$
- (xiii) If $y = \tan^{-1}\left(\frac{1-\cos x}{\sin x}\right)$, find $\frac{dy}{dx}$
- (xiv) If $y = \tan^{-1}\left(\frac{\cos x}{1-\sin x}\right)$, find $\frac{dy}{dx}$
42. If $y = \sin^{-1}\left(\frac{2x}{1+x^2}\right) + \cos^{-1}\left(\frac{1-x^2}{1+x^2}\right)$,
 $0 < x < 1$, prove that, $\frac{dy}{dx} = \frac{4}{1+x^2}$.
43. If $y = \sin^{-1}\left(\frac{x}{\sqrt{1+x^2}}\right) + \cos^{-1}\left(\frac{1}{\sqrt{x^2+1}}\right)$,
 $0 < x < \infty$, prove that, $\frac{dy}{dx} = \frac{2}{1+x^2}$
44. If $y = \tan^{-1}\left\{\frac{\sqrt{1+x^2} + \sqrt{1-x^2}}{\sqrt{1+x^2} - \sqrt{1-x^2}}\right\}$, find $\frac{dy}{dx}$.
45. If $y = \cot^{-1}\left\{\frac{\sqrt{1+\sin x} + \sqrt{1-\sin x}}{\sqrt{1+\sin x} - \sqrt{1-\sin x}}\right\}$,
 then prove that $\frac{dy}{dx}$ is independent of x .
46. If $y = \tan^{-1}\left(\frac{2x}{1-x^2}\right) + \sec^{-1}\left(\frac{1+x^2}{1-x^2}\right)$, $x > 0$,
 then prove that $\frac{dy}{dx} = \frac{4}{1+x^2}$.
47. If $y = \sec^{-1}\left(\frac{x-1}{x+1}\right) + \sin^{-1}\left(\frac{1+x^2}{1-x^2}\right)$, $x > 0$,
 prove that $\frac{dy}{dx} = 0$.
48. If $y = \sin\left[2\tan^{-1}\left\{\sqrt{\frac{1-x}{1+x}}\right\}\right]$, find $\frac{dy}{dx}$.
49. If $y = \cos^{-1}(2x) + 2\cos^{-1}(\sqrt{1-4x^2})$,
 $0 < x < \frac{1}{2}$, find $\frac{dy}{dx}$.
50. If $y = \tan^{-1}\left(\frac{\sqrt{1+x} - \sqrt{1-x}}{\sqrt{1+x} + \sqrt{1-x}}\right)$,
 find $\frac{dy}{dx}$.
51. If $y = \cos^{-1}x + \cos^{-1}\left(\frac{x}{2} + \frac{\sqrt{3-3x^2}}{2}\right)$

$$\frac{1}{2} < x < 1, \text{ find } \frac{dy}{dx}.$$

52. If $y = \sin^{-1}(x\sqrt{1-x} - \sqrt{x-x^3})$, find $\frac{dy}{dx}$
53. If $y = f(x) = x^3 + x^5 + x^7$ and g is the inverse of f , then find $g'(3)$
54. If $y = f(x) = x^5 + 2x^3 + 2x$ and g is the inverse of f , find $g'(-5)$
55. If $f(x) = x^3 + 2x^2 + 3x + 4$ and $g(x)$ is the inverse of $f(x)$, find $g'(4)$.
56. If f and g are inverse of each other and $f'(x) = \frac{1}{1+x^n}$ find $g'(x)$.

Differentiation of Implicit Functions

57. Find $\frac{dy}{dx}$, if $2x^2 + 3xy + 3y^2 = 1$
58. If $e^x + e^y = e^{x+y}$, prove that, $\frac{dy}{dx} + e^{y-x} = 0$
59. If $x\sqrt{1+y} + y\sqrt{1+x} = 0$, prove that $\frac{dy}{dx} = -\frac{1}{(1+x)^2}$
60. If $y\sqrt{1-x^2} + x\sqrt{1-y^2} = 1$ prove that

$$\frac{dy}{dx} = -\sqrt{\frac{1-y^2}{1-x^2}}$$
61. If $\sin y = x \sin(a+y)$, prove that $\frac{dy}{dx} = \frac{\sin^2(a+y)}{\sin a}$
62. If $\log(x+y) = 2xy$, then prove that $y'(0) = 1$.
63. If $\sqrt{1-x^6} + \sqrt{1-y^6} = a(x^3 - y^3)$ prove that

$$\frac{dy}{dx} = \frac{x^2}{y^2} \sqrt{\frac{1-y^6}{1-x^6}}$$
64. If $x^2 + y^2 = t - \frac{1}{t}$ and $x^4 + y^4 = t^2 + \frac{1}{t^2}$ then prove
 that $\frac{dy}{dx} = \frac{1}{x^3 y}$.
65. If $\sec\left(\frac{x+y}{x-y}\right) = a$, prove that $\frac{dy}{dx} = \frac{y}{x}$.
66. If $\tan^{-1}\left(\frac{x^2 - y^2}{x^2 + y^2}\right) = a$ prove that $\frac{dy}{dx} = \frac{x(1 - \tan a)}{y(1 + \tan a)}$.
67. If $xy = 1$, prove that $\frac{dy}{dx} + y^2 = 0$
68. If $y = x \sin y$, prove that $\frac{dy}{dx} = \frac{y}{x(1 - x \cos y)}$.
69. If $\cos y = x \cos(a+y)$ with $\cos a \neq \pm 1$,

prove that $\frac{dy}{dx} = \frac{\cos^2(a+y)}{\sin a}$

Logarithmic Differentiation

70. Find $\frac{dy}{dx}$, if $y = x^{\sin x}$
71. Find $\frac{dy}{dx}$, if $y = (\sin x)^{\cos x}$
72. Find $\frac{dy}{dx}$, if $y = (\tan x)^{\cot x} + (\cot x)^{\tan x}$
73. If $x^m y^n = (x+y)^{m+n}$, prove that, $\frac{dy}{dx} = \frac{y}{x}$
74. If $y = \left(1 + \frac{1}{x}\right)^x + x^{\left(1 + \frac{1}{x}\right)}$, find $\frac{dy}{dx}$ at $x = 1$
75. If $y = \sin x \cdot \sin 2x \cdot \sin 3x \dots \sin(2014)x$, find $\frac{dy}{dx}$
76. If $x^y = e^{x-y}$, prove that, $\frac{dy}{dx} = \frac{\log x}{(1 + \log x)^2}$
77. If $x^y = y^x$, find $\frac{dy}{dx}$
78. If $e^y = y^x$, prove that $\frac{dy}{dx} = \frac{(\log y)^2}{\log y - 1}$
79. If $x^m y^n = 1$, prove that, $\frac{dy}{dx} = -\frac{my}{nx}$
80. If $e^{x+y} - x = 0$, prove that $\frac{dy}{dx} = \frac{1-x}{x}$
81. Find $\frac{dy}{dx}$, if $y = x^{\sin x - \cos x} + \frac{x^2 - 1}{x^2 + 1}$
82. Find $y'(0)$, if
 $y = (x+1)(x+2)(x+3) \dots (x+2012)$
83. Find $y'(0)$, if
 $y = (1+x)(1+x^2)(1+x^4)(1+x^8) \dots (1+x^{1006})$

Differentiation of Infinite Series

84. If $y = \sqrt{\cos x} + \sqrt{\cos x} + \sqrt{\cos x} + \dots$ to ∞
then find $\frac{dy}{dx}$
85. If $y = \frac{\sin x}{1+} \frac{\cos x}{1+} \frac{\sin x}{1+} \frac{\cos x}{1+} \dots$, find $\frac{dy}{dx}$
86. If $y = \sqrt{x} + \sqrt{x} + \sqrt{x} + \dots$ to ∞ , prove that
 $\frac{dy}{dx} = \frac{1}{2y-1}$
87. If $y = x^{x \dots 1024}$, prove that $\frac{dy}{dx} = \frac{y^2}{x(1 - y \log x)}$
88. If $y = \sqrt{\sin x} + \sqrt{\sin x} + \sqrt{\sin x} + \dots$ to ∞ , prove that
 $\frac{dy}{dx} = \frac{\cos x}{2y-1}$
89. If $y = e^{x+e^{x+e^{x+\dots}}}$, prove that $\frac{dy}{dx} = \frac{y}{1-y}$
90. If $y = (\tan)^{(\tan)^{(\tan) \dots \text{to } x}}$, prove that $\frac{dy}{dx}$ at $x = \frac{\pi}{4}$
91. If $y = x + \frac{1}{x} + \frac{1}{x} + \frac{1}{x} + \dots$ to ∞ , then prove that,
 $\frac{dy}{dx} = \frac{y}{2y-x}$
92. If $y = \frac{\sin x}{1+} \frac{\cos x}{1+} \frac{\sin x}{1+} \frac{\cos x}{1+} \dots$ to ∞ ,
then prove that, $y'(0) = \frac{1}{2}$
93. If $y = \frac{x}{1+} \frac{x}{2+} \frac{x}{1+} \frac{x}{2+} \dots$ to ∞ , then find $\frac{dy}{dx}$

Differentiation of Parametric Functions

94. Find $\frac{dy}{dx}$, when
 $x = a(t - \sin t)$, $y = a(l - \cos t)$
95. If $x^2 - y^2 = t - \frac{1}{t}$ and $x^4 + y^4 = t^2 + \frac{1}{t^2}$
then prove that $x^3 y \frac{dy}{dx} + 1 = 0$
96. If $x = a\left(t + \frac{1}{t}\right)$ and $y = a\left(t - \frac{1}{t}\right)$,
then prove that $\frac{dy}{dx} = \frac{x}{y}$
97. If $y = \cos^{-1}\left(\frac{5t + 12\sqrt{1-t^2}}{13}\right)$
and $x = \cos^{-1}\left(\frac{1-t^2}{1+t^2}\right)$, find $\frac{dy}{dx}$
98. If $x = \sin^{-1}\left(\frac{3\sin t + 4\cos t}{5}\right)$
and $y = \sin^{-1}\left(\frac{6\cos t + 8\sin t}{10}\right)$, find $\frac{dy}{dx}$
99. If $x = \cos t$, and $y = \sin t$, prove that $\frac{dy}{dx} = \frac{1}{\sqrt{3}}$ at
 $x = \frac{2\pi}{3}$
100. If $x = e^{\cos 2t}$ and $y = e^{\sin 2t}$, prove that $\frac{dy}{dx} = \frac{y \log x}{x \log y}$
101. If $x = x = 2\cos \theta - \cos(2\theta)$ and
 $y = 2\sin \theta - \sin 2\theta$, prove that

$$\frac{dy}{dx} = \tan\left(\frac{3\theta}{2}\right)$$

102. If $x = a\left(t + \frac{1}{t}\right)$ and $y = a\left(t - \frac{1}{t}\right)$,

then prove that $\frac{dy}{dx} = \frac{x}{y}$

103. If $x = \sin^{-1}\left(\frac{2t}{1+t^2}\right)$ and $y = \tan^{-1}\left(\frac{2t}{1-t^2}\right)$,

$t > 1$, then prove that $\frac{dx}{dy} = -1$

104. If $y = \cos^{-1}\left(\frac{5t + 12\sqrt{1-t^2}}{13}\right)$

and $x = \cos^{-1}\left(\frac{1-t^2}{1+t^2}\right)$, find $\frac{dy}{dx}$.

105. If $x = \sin^{-1}\left(\frac{3\sin t + 4\cos t}{5}\right)$

and $y = \sin^{-1}\left(\frac{6\cos t + 8\sin t}{10}\right)$, find $\frac{dy}{dx}$.

106. If $x = \tan^{-1}\left(\frac{\sqrt{1+\sin t} + \sqrt{1-\sin t}}{\sqrt{1+\sin t} - \sqrt{1-\sin t}}\right)$

and $y = \tan^{-1}\left(\frac{\sqrt{1+t^2} - 1}{t}\right)$, find $\frac{dy}{dx}$.

Differentiation of a Function with Respect to Another Function

107. Differentiate $f(x^2 + 2012)$ w.r.t $f(x^3 + 2013)$.

108. Differentiate x^x w.r.t $x \log x$.

109. Differentiate $\tan^{-1}\left(\frac{\sqrt{1+x^2} - \sqrt{1-x^2}}{\sqrt{1+x^2} + \sqrt{1-x^2}}\right)$

w.r.t $\cos^{-1}x^2$.

110. Differentiate $\sin^{-1}\left(\frac{2x}{1+x^2}\right)$ w.r.t $\cos^{-1}\left(\frac{1-x^2}{1+x^2}\right)$

111. Differentiate $f(\sin x)$ w.r.t $f(\cos x)$

Higher Order Derivatives

112. If $y = c_1 e^x + c_2 e^{-x}$, then prove that $\frac{d^2y}{dx^2} - y = 0$.

113. If $x = at^2$, $y = 2at$, find $\frac{d^2y}{dx^2}$

114. If $x = a \cos \theta$, $y = b \sin \theta$, find $\frac{d^2y}{dx^2}$

115. Prove that $\frac{d^2y}{dx^2} = -\left(\frac{d^2y}{dx^2}\right)\left(\frac{dy}{dx}\right)^{-3}$

116. If $y = e^{2x}$, find $\left(\frac{d^2y}{dx^2}\right)\left(\frac{d^2x}{dy^2}\right)$.

117. If $(a + bx)e^{\frac{x}{y}} = x$, then prove that $x^3 \frac{d^2y}{dx^2} = \left(x \frac{dy}{dx} - y\right)^2$.

118. If $x^2 + y^2 = a^2$ and $k = \frac{1}{a}$, then prove that $k = \frac{|y''|}{\sqrt{(1+y'^2)^3}}$.

119. If $y = e^{ax} \sin bx$, prove that, $y^2 - 2ay_1 + (a^2 + b^2)y = 0$

120. If $y = 2 \sin x + 3 \cos x$, prove that, $\frac{d^2y}{dx^2} + y = 0$

121. If $y = x + \tan x$, prove that $\cos^2 x \frac{d^2y}{dx^2} - 2y + 2x = 0$

122. If $y = \tan x + \sec x$, prove that $\frac{d^2y}{dx^2} = \frac{\cos x}{(1 - \sin x)^2}$

123. If $y = A \cos(\log x) + B \sin(\log x)$, then prove that

$$x^2 \frac{d^2y}{dx^2} + x \frac{dy}{dx} + y = 0$$

124. If $y = \tan^{-1} x$, prove that $(1 + x^2) y_2 + 2xy_1 = 0$

125. If $y = e^x (\sin x + \cos x)$, prove that $y_2 - 2y_1 + 2y = 0$

126. If $y = \sin^{-1} x$, prove that $(1 - x^2) y_2 - xy_1 = 0$

127. Find $\frac{d^2y}{dx^2}$, if

(i) $x = at^2$, $y = 2at$

(ii) $x = a \cos^3 \theta$, $y = a \sin^3 \theta$

(iii) $x = a \cos \theta$, $y = b \sin \theta$

128. If $x = a \sec \theta$, $y = b \tan \theta$, prove that $\frac{d^2y}{dx^2} = -\frac{b^4}{a^2 y^3}$.

129. If $y = \sin^{-1} x$, prove that $(1 - x^2) y_2 - xy_1 = 0$

130. Find $\frac{d^2y}{dx^2}$, if

(i) $x = at^2$, $y = 2at$

(ii) $x = a \cos^3 \theta$, $y = a \sin^3 \theta$

(iii) $x = a \cos \theta$, $y = b \sin \theta$

131. If $x = a \sec \theta$, $y = b \tan \theta$ prove that $\frac{d^2y}{dx^2} = -\frac{b^4}{a^2 y^3}$.

132. If $x = a(\theta + \sin \theta)$, $y = a(1 + \cos \theta)$

prove that $\frac{d^2y}{dx^2} = -\frac{a}{y^2}$

133. If $x = a(1 - \cos \theta)$, $y = a(\theta + \sin \theta)$, prove that

$$\frac{d^2y}{dx^2} = -\frac{1}{a} \text{ at } \theta = \frac{\pi}{2}.$$

134. If $y = x \log \left(\frac{x}{a+bx} \right)$, prove that $x^3 \frac{d^2y}{dx^2} = \left(x \frac{dy}{dx} - y \right)^2$.

135. If $\sqrt{y+x} + \sqrt{y-x} = c$, then prove that $\frac{d^2y}{dx^2} = \frac{2}{c^2}$.

136. If $\frac{\cos^4 \alpha}{x} + \frac{\sin^4 \alpha}{y} = \frac{1}{x+y}$, then prove that $\frac{dy}{dx} = \tan^2 \alpha$.

137. If $y = x \sin x$, then prove that,

$$x^2 \frac{d^2y}{dx^2} - 2x \frac{dy}{dx} + (x^2 + 2)y = 0.$$

138. If $f: R \rightarrow R$ be a function is defined by

$$y = f(x) = x^2, \text{ then prove that, } \left(\frac{d^2y}{dx^2} \right) \times \left(\frac{d^2y}{dx^2} \right) = -\frac{1}{2x^3}.$$

139. There is a polynomial

$$P(x) = ax^3 + bx^2 + cx + d \text{ such that}$$

$$P(0) = P(1) = -2, P'(0) = -1, \text{ then find the value of } a + b + c + d + 10.$$

140. If $f(x) = x + \tan x$ and g is the inverse of f , then

$$\text{prove that } g'(x) = \frac{1}{2 + \tan^2(g(x))}.$$

141. If g is the inverse of f such that $f(x) = e^x + x^3 - 1$,

$$\text{then prove that } g''(e) = -\frac{e+6}{(e+3)^3}.$$

142. Let $f(x) = 1 + x^3$. If $g(x) = f^{-1}(x)$, then prove that

$$g'''(2) = \frac{8}{3}.$$

143. Suppose f is a differentiable function such that

$$f(g(x)) = x \text{ and } f'(x) = 1 + (f(x))^2, \text{ then prove that}$$

$$g'(x) = \frac{1}{1 + x^2}.$$

Differentiation of a Determinant

144. Let $f(x) = \begin{vmatrix} x^3 & \sin x & \cos x \\ 6 & -1 & 0 \\ p & p^2 & p^3 \end{vmatrix}$, where p is a constant

$$\text{then prove that } \frac{d^3}{dx^3} (f(x)) \text{ at } x = 0 \text{ is } 0.$$

145. Find the coefficient of x in the determinant

$$\begin{vmatrix} (1+x)^{a_1b_1} & (1+x)^{a_1b_2} & (1+x)^{a_1b_3} \\ (1+x)^{a_2b_1} & (1+x)^{a_2b_2} & (1+x)^{a_2b_3} \\ (1+x)^{a_3b_1} & (1+x)^{a_3b_2} & (1+x)^{a_3b_3} \end{vmatrix}$$

with the help of determinant.

146. Let $f(x) = \begin{vmatrix} x+1 & x+2 & x+3 \\ x+4 & x+5 & x+6 \\ x+7 & x+8 & x+9 \end{vmatrix}$

Then prove that $f(x)$ is a linear polynomial of x .

147. Let $f(x) = \begin{vmatrix} 1 & 2x & 3x^2 \\ x & x^2 & x^2 \\ 0 & 2 & 6x \end{vmatrix}$

Then find the value of $f'(1)$.

148. Let $f(x) = \begin{vmatrix} \cos x & \sin x & \cos x \\ \cos 2x & \sin 2x & 2 \cos 2x \\ \cos 3x & \sin 3x & 2 \cos 3x \end{vmatrix}$

Then find the value of $f'\left(\frac{\pi}{2}\right)$.

Leibnitz Rules for Differentiation

149. If $y = e^x$, find the value of $y_n(0)$.

150. If $y = a^x$, find the value of $y_n(0)$.

151. If $y = \frac{1}{(x+1)}$, find the value of $y_n(0)$.

152. If $y = \sin x$, find the value of $y_n(0)$.

153. If $y = x^n$, find the value of $y_n(1)$.

154. If $y = A \sin x + B \cos x$, then prove that

(i) $y_2 + y = 0$

(ii) $y_{n+2} + y_n = 0$.

155. If $y = \tan^{-1} x$, then prove that

(i) $(1+x^2)y_2 + 2xy_1 = 0$

(ii) $(1+x^2)y_{n+2} + 2(n+2)xy_{n+1} + n(n+1)y_n = 0$.

156. If $y = \sin^{-1} x$, then prove that

(i) $(1-x^2)y_2 - xy_1 = 0$

(ii) $(1-x^2)y_{n+2} - (2n+1)xy_{n+1} - n^2y_n = 0$

157. If $y = a \cos(\log x) + b \sin(\log x)$, then prove that

$$x^2 \cdot y_{n+2} + (2n+3)x \cdot y_{n+1} + (n^2+1)y_n = 0.$$

Hints & Solutions

Level - 1

1. We have $f'(x)$

$$\begin{aligned}
 &= \lim_{h \rightarrow 0} \left(\frac{f(x+h) - f(x)}{h} \right) \\
 &= \lim_{h \rightarrow 0} \left(\frac{e^{x+h} - e^x}{h} \right) \\
 &= \lim_{h \rightarrow 0} \left(e^x \cdot \frac{e^h - 1}{h} \right) \\
 &= e^x
 \end{aligned}$$

2. We have $f'(x)$

$$\begin{aligned}
 &= \lim_{h \rightarrow 0} \left(\frac{f(x+h) - f(x)}{h} \right) \\
 &= \lim_{h \rightarrow 0} \left(\frac{\log(x+h) - \log x}{h} \right) \\
 &= \lim_{h \rightarrow 0} \left(\frac{\log \left(\frac{x+h}{x} \right)}{h} \right) \\
 &= \lim_{h \rightarrow 0} \left(\frac{\log \left(1 + \frac{h}{x} \right)}{h} \right) \\
 &= \lim_{h \rightarrow 0} \left(\frac{\log \left(1 + \frac{h}{x} \right)}{\frac{h}{x} \cdot x} \right) \\
 &= \lim_{h \rightarrow 0} \left(\frac{1}{x} \left(\frac{\log \left(1 + \frac{h}{x} \right)}{\frac{h}{x}} \right) \right) \\
 &= 1/x
 \end{aligned}$$

3. We have $f'(x)$

$$\begin{aligned}
 &= \lim_{h \rightarrow 0} \left(\frac{f(x+h) - f(x)}{h} \right) \\
 &= \lim_{h \rightarrow 0} \left(\frac{\log_a(x+h) - \log_a x}{h} \right) \\
 &= \lim_{h \rightarrow 0} \left(\frac{\log_a \left(\frac{x+h}{x} \right)}{h} \right) \\
 &= \lim_{h \rightarrow 0} \left(\frac{\log_a \left(1 + \frac{h}{x} \right)}{\frac{h}{x} \cdot x} \right) \\
 &= \lim_{h \rightarrow 0} \frac{1}{x} \left(\frac{\log_e \left(1 + \frac{h}{x} \right)}{\frac{h}{x} \cdot \log_e a} \right) \\
 &= \lim_{h \rightarrow 0} \left(\frac{1}{x \cdot \log_e a} \left(\frac{\log_e \left(1 + \frac{h}{x} \right)}{\frac{h}{x}} \right) \right) \\
 &= \frac{1}{x \cdot \log_e a}
 \end{aligned}$$

4. We have $f'(x)$

$$\begin{aligned}
 &= \lim_{h \rightarrow 0} \left(\frac{f(x+h) - f(x)}{h} \right) \\
 &= \lim_{h \rightarrow 0} \left(\frac{\sin(x+h) - \sin x}{h} \right) \\
 &= \lim_{h \rightarrow 0} \left(\frac{2 \cos \left(\frac{x+h+x}{2} \right) \sin \left(\frac{x+h-x}{2} \right)}{h} \right) \\
 &= \lim_{h \rightarrow 0} \left(\frac{2 \cos \left(x + \frac{h}{2} \right) \sin \left(\frac{h}{2} \right)}{h} \right) \\
 &= \lim_{h \rightarrow 0} \left(\frac{2 \cos \left(x + \frac{h}{2} \right) \sin \left(\frac{h}{2} \right)}{\frac{h}{2} \cdot 2} \right) \\
 &= \cos x
 \end{aligned}$$

5. We have $f'(x)$

$$\begin{aligned}
 &= \lim_{h \rightarrow 0} \left(\frac{f(x+h) - f(x)}{h} \right) \\
 &= \lim_{h \rightarrow 0} \left(\frac{\tan^{-1}(x+h) - \tan^{-1}x}{h} \right) \\
 &= \lim_{h \rightarrow 0} \left(\frac{\tan^{-1} \left(\frac{x+h-x}{1+x(x+h)} \right)}{h} \right) \\
 &= \lim_{h \rightarrow 0} \left(\frac{\tan^{-1} \left(\frac{h}{1+x(x+h)} \right)}{h} \right) \\
 &= \lim_{h \rightarrow 0} \left(\frac{\tan^{-1} \left(\frac{h}{1+x(x+h)} \right)}{\frac{h}{1+x(x+h)} \cdot (1+x(x+h))} \right) \\
 &= \frac{1}{1+x^2}
 \end{aligned}$$

6. We have $f'(x)$

$$\begin{aligned}
 &= \lim_{h \rightarrow 0} \left(\frac{f(x+h) - f(x)}{h} \right) \\
 &= \lim_{h \rightarrow 0} \left(\frac{e^{\sin(x+h)} - e^{\sin x}}{h} \right) \\
 &= \lim_{h \rightarrow 0} \left(\frac{e^{\sin(x+h)} (e^{\sin(x+h) - \sin x} - 1)}{h} \right) \\
 &= \lim_{h \rightarrow 0} \left(\frac{e^{\sin(x+h)} (e^{\sin(x+h) - \sin x} - 1)}{(\sin(x+h) - \sin x)} \times \frac{\sin(x+h) - \sin x}{h} \right) \\
 &= \lim_{h \rightarrow 0} \left(\frac{e^{\sin x} \times (\sin(x+h) - \sin x)}{h} \right)
 \end{aligned}$$

$$= (e^{\sin x} \times \cos x)$$

7. We have $f'(x)$

$$\begin{aligned}
 &= \lim_{h \rightarrow 0} \left(\frac{f(x+h) - f(x)}{h} \right) \\
 &= \lim_{h \rightarrow 0} \left(\frac{\sin^2(x+h) - \sin^2(x)}{h} \right) \\
 &= \lim_{h \rightarrow 0} \left(\frac{\cos \left(\frac{(x+h)^2 + x^2}{2} \right) \sin \left(\frac{(x+h)^2 - x^2}{2} \right)}{h} \right) \\
 &= \lim_{h \rightarrow 0} \left(\frac{2 \cos \left(\frac{(x+h)^2 + x^2}{2} \right) \sin \left(\frac{(2hx + h^2)}{2} \right)}{\left(\frac{2hx + h^2}{2} \right) \left(\frac{2h}{2hx + h^2} \right)} \right) \\
 &= \lim_{h \rightarrow 0} \left(\frac{2 \cos \left(\frac{(x+h)^2 + x^2}{2} \right) \sin \left(\frac{(2hx + h^2)}{2} \right)}{\left(\frac{2hx + h^2}{2} \right) \left(\frac{2}{2x+h} \right)} \right) \\
 &= \lim_{h \rightarrow 0} \left(\frac{\cos \left(\frac{(x+h)^2 + x^2}{2} \right)}{\left(\frac{1}{2x+h} \right)} \right) \\
 &= 2x \cos(x^2).
 \end{aligned}$$

8. We have $y = \log_x x + 10 = 1 + 10 = 11$

$$\Rightarrow \frac{dy}{dx} = 0.$$

9. We have $y = 5^{\log_3 x} - x^{\log_3 5}$

$$\Rightarrow y = x^{\log_3 x} - x^{\log_3 5} = 0$$

$$\Rightarrow \frac{dy}{dx} = 0$$

10. We have $y = \frac{1}{x} + \frac{2}{x^2} + \frac{3}{x^3}$

$$\Rightarrow \frac{dy}{dx} = -\frac{1}{x^2} - \frac{4}{x^3} - \frac{9}{x^4}$$

11. We have $y = \sqrt{x} + x\sqrt{x} + x^2\sqrt{x} + x^3\sqrt{x}$

$$= x^{\frac{1}{2}} + x^{\frac{3}{2}} + x^{\frac{5}{2}} + x^{\frac{7}{2}}$$

$$\Rightarrow \frac{dy}{dx} = \frac{1}{2\sqrt{x}} + \frac{3}{2}\sqrt{x} + \frac{5}{2}x\sqrt{x} + \frac{7}{2}x^2\sqrt{x}$$

12. We have

$$y = \left(1 + \tan \left(\frac{\pi}{8} - x \right) \right) \left(1 + \tan \left(x + \frac{\pi}{8} \right) \right)$$

$$\Rightarrow y = 2$$

$$\Rightarrow \frac{dy}{dx} = 0$$

$$\begin{aligned} 13. \text{ We have } y &= \frac{\sec x + \tan x - 1}{\sec x - \tan x + 1} \\ &= \frac{\sec x + \tan x - (\sec^2 x - \tan^2 x)}{\sec x - \tan x + 1} \\ &= \frac{(\sec x + \tan x)(1 + \sec x - \tan x)}{(\sec x - \tan x + 1)} \\ &= (\sec x + \tan x) \end{aligned}$$

$$\text{Thus, } \frac{dy}{dx} = (\sec x \tan x + \sec^2 x)$$

$$\left(\frac{dy}{dx}\right)_{x=0} = 0 + 1 = 1$$

$$\begin{aligned} 14. \text{ We have } y &= \frac{x^4 + x^2 + 1}{x^2 - x + 1} \\ &= \frac{(x^2 - x + 1)(x^2 + x + 1)}{(x^2 - x + 1)} \\ &= (x^2 + x + 1) \end{aligned}$$

$$\text{Thus, } \frac{dy}{dx} = 2x + 1$$

$$\text{Given } \frac{dy}{dx} = ax + b$$

Comparing the co-efficients of x and the constant term, we get, $a = 2$ and $b = 1$

$$\text{Thus, } a + b + 10 = 2 + 1 + 10 = 13.$$

15.

16.

$$17. \text{ Given } y = \log(\sin(3x + 5))$$

$$\Rightarrow \frac{dy}{dx} = \frac{1}{\sin(3x + 5)} \cdot \cos(3x + 5) \cdot 3$$

$$\Rightarrow \frac{dy}{dx} = 3 \cot(3x + 5)$$

$$18. \text{ Given } y = \log(x + \sqrt{x^2 + 1})$$

$$\Rightarrow \frac{dy}{dx} = \frac{1}{(x + \sqrt{x^2 + 1})} \times \left(1 + \frac{1 \times 2x}{2\sqrt{x^2 + 1}}\right)$$

$$\Rightarrow \frac{dy}{dx} = \frac{1}{(x + \sqrt{x^2 + 1})} \times \left(1 + \frac{x}{\sqrt{x^2 + 1}}\right)$$

$$\Rightarrow \frac{dy}{dx} = \frac{1}{(x + \sqrt{x^2 + 1})} \times \left(\frac{\sqrt{x^2 + 1} + x}{\sqrt{x^2 + 1}}\right)$$

$$\Rightarrow \frac{dy}{dx} = \frac{1}{\sqrt{x^2 + 1}}$$

$$19. \text{ Given } y = \log(\sqrt{x-1} + \sqrt{x+1})$$

$$\Rightarrow \frac{dy}{dx} = \frac{1}{(\sqrt{x-1} + \sqrt{x+1})} \times \left(\frac{1}{2\sqrt{x-1}} - \frac{1}{2\sqrt{x+1}}\right)$$

$$\Rightarrow \frac{dy}{dx} = \frac{1}{(\sqrt{x-1} + \sqrt{x+1})} \times \left(\frac{\sqrt{x+1} - \sqrt{x-1}}{2\sqrt{x^2-1}}\right)$$

$$\Rightarrow \frac{dy}{dx} = -\frac{1}{2\sqrt{x^2-1}}$$

$$20. \text{ Given } y = (\sin x)^x$$

$$\Rightarrow y = e^{\log(\sin x)^x} = e^{x \log(\sin x)}$$

$$\Rightarrow \frac{dy}{dx} = e^{x \log(\sin x)} \left(\log \sin x + x \cdot \frac{\cos x}{\sin x}\right)$$

$$\Rightarrow \frac{dy}{dx} = (\sin x)^x (\log \sin x + x \cdot \cot x)$$

$$21. \text{ Given } y = \log(\sin x + \cos x)$$

$$\Rightarrow \frac{dy}{dx} = \frac{\cos x - \sin x}{\sin x + \cos x}$$

$$\Rightarrow \frac{dy}{dx} = \frac{1 - \tan x}{1 + \tan x}$$

$$\Rightarrow \frac{dy}{dx} = \tan\left(\frac{\pi}{4} - x\right)$$

$$22. \text{ We have } f(x) = \frac{1}{x-1}$$

$$\text{Now, } f(f(x)) = f\left(\frac{1}{x-1}\right)$$

$$= \frac{1}{\frac{1}{x-1} - 1}$$

$$= \frac{x-1}{1-x+1}$$

$$= \frac{x-1}{2-x}$$

$$\text{Thus, } \frac{d((f(f(x))))}{dx}$$

$$= \frac{d}{dx} \left(\frac{x-1}{2-x}\right)$$

$$= \frac{(2-x) \cdot 1 - (x-1) \cdot (-1)}{(2-x)^2}$$

$$= \frac{(2-x+x-1)}{(2-x)^2} = \frac{1}{(2-x)^2}$$

$$23. \text{ Given } f(x) = \frac{x-1}{x+1}$$

$$\begin{aligned}
 f(f(f(x))) &= f\left(f\left(\frac{x-1}{x+1}\right)\right) \\
 &= f\left(\frac{\left(\frac{x-1}{x+1}\right) - 1}{\left(\frac{x-1}{x+1}\right) + 1}\right) \\
 &= f\left(\frac{x-1-x-1}{x-1+x+1}\right) \\
 &= f\left(\frac{-2}{2x}\right) \\
 &= f\left(\frac{-1}{x}\right) \\
 &= \frac{\left(\frac{-1}{x} - 1\right)}{\left(\frac{-1}{x} + 1\right)} \\
 &= \frac{(-1-x)}{(1-x)} = \frac{(x+1)}{(x-1)}
 \end{aligned}$$

Thus, $\frac{d(f(f(f(x))))}{dx}$

$$\begin{aligned}
 &= \frac{d}{dx} \left(\frac{x+1}{x-1} \right) \\
 &= \frac{(x-1) \cdot 1 - (x+1) \cdot 1}{(x-1)^2} \\
 &= \frac{(x-1-x-1)}{(x-1)^2} = \frac{-2}{(x-1)^2}
 \end{aligned}$$

24. Given $g(x) = f(3x-1)$

$$\begin{aligned}
 g'(x) &= 3f'(3x-1) \\
 &= 3 \times \frac{1}{(3x-1)^2 + 1} \\
 &= \frac{3}{9x^2 - 6x + 2}
 \end{aligned}$$

25. Given $y = f(\sin(x^3))$

$$\begin{aligned}
 \Rightarrow \frac{dy}{dx} &= f'(\sin(x^3)) \cdot \cos(x^3) \cdot 3x^2 \\
 \Rightarrow \frac{dy}{dx} &= 3x^2 \cdot \cos(x^3) \cdot ((\sin(x^3))^2 + 1)
 \end{aligned}$$

26. Given $f(x) = |x-1| + |x-3|$

$$\begin{aligned}
 &= \begin{cases} -(x-1) - (x-3) & : x < 1 \\ (x-1) - (x-3) & : 1 \leq x < 3 \\ (x-1) + (x-3) & : x \geq 3 \end{cases} \\
 &= \begin{cases} -2x+2 & : x < 1 \\ 2 & : 1 \leq x < 3 \\ 2x-4 & : x \geq 3 \end{cases}
 \end{aligned}$$

Thus, $f'(2) = 0$

27. Given $f(x) = |x^2-1| + |x^2-4|$

$$= \begin{cases} -(x^2-1) - (x^2-4) & : x < -2 \\ (x^2-1) - (x^2-4) & : -2 \leq x < -1 \\ (1-x^2) + (4-x^2) & : -1 \leq x < 1 \\ (x^2-1) + (4-x^2) & : 1 \leq x < 2 \\ (x^2-1) + (x^2-4) & : x \geq 2 \end{cases}$$

Thus, $f'\left(\frac{3}{2}\right) = 0$

28. Given $f(x^2) = x^4 + x^3 + 1$

$$\begin{aligned}
 \Rightarrow f'(x^2) \cdot 2x &= 4x^3 + 3x^2 \\
 \Rightarrow f'(x^2) &= \frac{4x^3 + 3x^2}{2x} = 2x^2 + \frac{3}{2}x
 \end{aligned}$$

Replacing x by x^2 , we get,

$$f'(x^4) = 2x^4 + \frac{3}{2}x^2$$

29. We have $g(x) = f^{-1}(x)$

$$\begin{aligned}
 \Rightarrow f(g(x)) &= x \\
 \Rightarrow f(g(x)) \cdot g'(x) &= 1 \\
 \Rightarrow g'(x) &= \frac{1}{f'(g(x))} \\
 \Rightarrow g'(x) &= \frac{1}{\cos(2g(x))}
 \end{aligned}$$

30. We have $f(x) = x + \tan x$

Given $f(x) = g^{-1}(x)$

$$\begin{aligned}
 \Rightarrow g(f(x)) &= x \\
 \Rightarrow g'(f(x)) \cdot f'(x) &= 1 \\
 \Rightarrow g'(f(x)) &= \frac{1}{f'(x)} = \frac{1}{1 + \sec^2 x} \\
 \Rightarrow g'(f(x)) &= \frac{1}{2 + \tan^2 x} \\
 \Rightarrow g'(f(x)) &= \frac{1}{2 + (f(x) - x)^2} \\
 \Rightarrow g'(x) &= \frac{1}{2 + (g(x) - x)^2}
 \end{aligned}$$

31. Given $y = \frac{x^4 - x^2 + 1}{x^2 + \sqrt{3x} + 1}$

$$= \frac{(x^4 + 1) - x^2}{x^2 + \sqrt{3x} + 1}$$

$$= \frac{(x^2 + 1)^2 - 3x^2}{x^2 + \sqrt{3}x + 1}$$

$$= \frac{(x^2 + \sqrt{3}x + 1)(x^2 - \sqrt{3}x + 1)}{(x^2 + \sqrt{3}x + 1)}$$

$$= (x^2 - \sqrt{3}x + 1)$$

$$\frac{dy}{dx} = 2x - \sqrt{3}$$

Given $\frac{dy}{dx} = px + q$

Comparing the co-efficients of x and constant term, we get, $p = 2$, $q = -\sqrt{3}$

Thus, $p - q$

$$= 2 + \sqrt{3}$$

$$= \cot\left(\frac{\pi}{12}\right)$$

$$= \tan\left(\frac{\pi}{2} - \frac{\pi}{12}\right)$$

$$= \tan\left(\frac{5\pi}{12}\right)$$

32. We have $y = \sqrt{x-1} + \sqrt{x+1}$

$$\frac{dy}{dx} = \frac{1}{2\sqrt{x-1}} + \frac{1}{2\sqrt{x+1}}$$

$$\frac{dy}{dx} = \frac{\sqrt{x+1} + \sqrt{x-1}}{2\sqrt{x-1}\sqrt{x+1}}$$

$$\frac{dy}{dx} = \frac{y}{2\sqrt{x^2-1}}$$

$$\sqrt{x^2-1} \frac{dy}{dx} = \frac{y}{2}$$

Hence, the result.

33. Given $y = \frac{x}{x+2} = \left(1 - \frac{1}{(x+2)}\right)$

$$\frac{dy}{dx} = \frac{1}{(x+2)^2}$$

$$x \frac{dy}{dx} = \frac{x}{(x+2)^2}$$

$$x \frac{dy}{dx} = \frac{1}{(x+2)} \cdot \frac{x}{(x+2)}$$

$$x \frac{dy}{dx} = \left(1 - \frac{x}{(x+2)}\right) \cdot \frac{x}{(x+2)}$$

$$x \frac{dy}{dx} = (1 - y)y$$

34. Given $y = \sqrt{x} + \frac{1}{\sqrt{x}}$

$$\frac{dy}{dx} = \frac{1}{2\sqrt{x}} - \frac{1}{2x\sqrt{x}}$$

$$2x \frac{dy}{dx} = \frac{2x}{2\sqrt{x}} - \frac{2x}{2x\sqrt{x}}$$

$$2x \frac{dy}{dx} = \sqrt{x} - \frac{1}{\sqrt{x}}$$

Hence, the result.

35. Given $y = e^x \cos x$

$$\frac{dy}{dx} = e^x \cos x - e^x \sin x$$

$$\frac{dy}{dx} = e^x (\cos x - \sin x)$$

$$\frac{dy}{dx} = \sqrt{2}e^x \left(\frac{1}{\sqrt{2}} \cos x - \frac{1}{\sqrt{2}} \sin x \right)$$

$$\frac{dy}{dx} = \sqrt{2}e^x \cos\left(x + \frac{\pi}{4}\right)$$

36. Given $y = \sqrt{x^2 + a^2}$

$$\frac{dy}{dx} = \frac{1 \times 2x}{2\sqrt{x^2 + a^2}}$$

$$\frac{dy}{dx} = \frac{x}{\sqrt{x^2 + a^2}}$$

$$\sqrt{x^2 + a^2} \frac{dy}{dx} = x$$

$$y \frac{dy}{dx} - x = 0$$

37. Given $y = e^x + e^{-x}$

$$\frac{dy}{dx} = e^x - e^{-x}$$

$$\frac{dy}{dx} = \sqrt{(e^x - e^{-x})^2}$$

$$\frac{dy}{dx} = \sqrt{(e^x - e^{-x})^2} - 4$$

$$\frac{dy}{dx} = \sqrt{y^2 - 4}$$

38. Given $xy = 4$

$$y = \frac{4}{x}$$

$$\frac{dy}{dx} = -\frac{4}{x^2}$$

$$\frac{dy}{dx} + y^2 = -\frac{4}{x^2} + y^2$$

$$\frac{dy}{dx} + y^2 = -\frac{4}{x^2} + \frac{16}{x^2} = \frac{12}{x^2}$$

$$x \left(\frac{dy}{dx} + y^2 \right) = x \times \frac{12}{x^2} = \frac{12}{x}$$

$$x\left(\frac{dy}{dx} + y^2\right) = 3\left(\frac{4}{x}\right) = 3y$$

$$39. \text{ Let } y = \frac{x}{2} \sqrt{a^2 - x^2} + \frac{a^2}{2} \sin^{-1}\left(\frac{x}{a}\right)$$

$$\begin{aligned} \frac{dy}{dx} &= \frac{1}{2} \sqrt{a^2 - x^2} + \frac{x}{2} \times \frac{1 \times -2x}{2\sqrt{a^2 - x^2}} \\ &+ \frac{a^2}{2} \times \frac{1}{\sqrt{1 - \frac{x^2}{a^2}}} \times \frac{1}{a} \\ &= \frac{1}{2} \sqrt{a^2 - x^2} - \frac{1}{2} \times \frac{x^2}{\sqrt{a^2 - x^2}} + \frac{a^2}{2} \times \frac{1}{\sqrt{a^2 - x^2}} \\ &= \frac{a^2 - 2x^2 + a^2}{2\sqrt{a^2 - x^2}} \\ &= \frac{2(a^2 - x^2)}{2\sqrt{a^2 - x^2}} \\ &= \sqrt{a^2 - x^2} \end{aligned}$$

Hence, the result.

40.

$$(i) \quad y = \tan^{-1}(\sqrt{1+x^2} + x)$$

Put $x = \tan \theta$

$$y = \tan^{-1}(\sec \theta + \tan \theta)$$

$$y = \tan^{-1}\left(\frac{1 + \sin \theta}{\cos \theta}\right)$$

$$y = \tan^{-1}\left(\frac{1 + \cos\left(\frac{\pi}{2} - \theta\right)}{\sin\left(\frac{\pi}{2} - \theta\right)}\right)$$

$$y = \tan^{-1}\left(\frac{2\cos^2\left(\frac{\pi}{4} - \frac{\theta}{2}\right)}{2\sin\left(\frac{\pi}{4} - \frac{\theta}{2}\right)\cos\left(\frac{\pi}{4} - \frac{\theta}{2}\right)}\right)$$

$$y = \tan^{-1}\left(\cot\left(\frac{\pi}{4} - \frac{\theta}{2}\right)\right)$$

$$y = \tan^{-1}\left(\tan\left(\frac{\pi}{2} - \left(\frac{\pi}{4} - \frac{\theta}{2}\right)\right)\right)$$

$$y = \left(\frac{\pi}{4} + \frac{\theta}{2}\right) = \frac{\pi}{4} + \frac{\tan^{-1} x}{2}$$

$$\frac{dy}{dx} = \frac{1}{2(1+x^2)}$$

$$(ii) \quad y = \tan^{-1}(\sqrt{1+x^2} - x)$$

Put $x = \tan \theta$

$$y = \tan^{-1}(\sec \theta - \tan \theta)$$

$$y = \tan^{-1}\left(\frac{1 - \sin \theta}{\cos \theta}\right)$$

$$y = \tan^{-1}\left(\frac{1 - \cos\left(\frac{\pi}{2} - \theta\right)}{\sin\left(\frac{\pi}{2} - \theta\right)}\right)$$

$$y = \tan^{-1}\left(\frac{2\sin^2\left(\frac{\pi}{2} - \frac{\theta}{2}\right)}{2\sin\left(\frac{\pi}{2} - \frac{\theta}{2}\right)\cos\left(\frac{\pi}{4} - \frac{\theta}{2}\right)}\right)$$

$$y = \tan^{-1}\left(\tan\left(\frac{\pi}{2} - \frac{\theta}{2}\right)\right)$$

$$y = \left(\frac{\pi}{2} - \frac{\theta}{2}\right) = \frac{\pi}{4} - \frac{\tan^{-1} x}{2}$$

$$\frac{dy}{dx} = -\frac{1}{2(1+x^2)}$$

(iii)

$$y = \tan^{-1}\left\{\frac{\sqrt{1+x^2} - 1}{x}\right\}, x \neq 0$$

Put

$$x = \tan \theta$$

$$y = \tan^{-1}\left(\frac{\sec \theta - 1}{\tan \theta}\right)$$

$$y = \tan^{-1}\left(\frac{1 - \cos \theta}{\sin \theta}\right)$$

$$y = \tan^{-1}\left(\frac{2\sin^2(\theta/2)}{2\sin(\theta/2)\cos(\theta/2)}\right)$$

$$y = \tan^{-1}(\tan(\theta/2)) = \frac{\theta}{2}$$

$$y = \frac{\tan^{-1} x}{2}$$

$$\frac{dy}{dx} = \frac{1}{2(1+x^2)}$$

(iv)

$$y = \tan^{-1}\left\{\frac{\sqrt{1+x^2} + 1}{x}\right\}, x \neq 0$$

Put

$$x = \tan \theta$$

$$y = \tan^{-1}\left(\frac{\sec \theta + 1}{\tan \theta}\right)$$

$$y = \tan^{-1}\left(\frac{1 + \cos \theta}{\sin \theta}\right)$$

$$y = \tan^{-1}\left(\frac{2\cos^2(\theta/2)}{2\sin(\theta/2)\cos(\theta/2)}\right)$$

$$y = \tan^{-1}(2\cos(\theta/2))$$

$$y = \tan^{-1}\left(\tan\left(\frac{\pi}{2} - \frac{\theta}{2}\right)\right)$$

$$y = \left(\frac{\pi}{2} - \frac{\theta}{2}\right)$$

$$y = \left(\frac{\pi}{2} - \frac{\tan^{-1} x}{2}\right)$$

$$\frac{dy}{dx} = -\frac{1}{2(1+x^2)}$$

(v)

$$y = \cot^{-1}(\sqrt{1+x^2} + x)$$

Put

$$x = \tan \theta$$

$$y = \cot^{-1}\left(\frac{\sec \theta + 1}{\tan \theta}\right)$$

$$y = \cot^{-1}\left(\frac{1 + \cos \theta}{\sin \theta}\right)$$

$$y = \cot^{-1} \left(\frac{2 \cos^2(\theta/2)}{2 \sin(\theta/2) \cos(\theta/2)} \right)$$

$$y = \cot^{-1}(\cot(\theta/2))$$

$$y = \frac{\theta}{2} = \frac{\tan^{-1} x}{2}$$

$$\frac{dy}{dx} = \frac{1}{2(1+x^2)}$$

$$(vi) \quad y = \tan^{-1} \left\{ \frac{\sqrt{1+\sin x} + \sqrt{1-\sin x}}{\sqrt{1+\sin x} - \sqrt{1-\sin x}} \right\}, \quad 0 < x < \pi$$

$$y = \tan^{-1} \left(\frac{\cos(x/2)}{\sin(x/2)} \right)$$

$$y = \tan^{-1} \left(\cot \left(\frac{x}{2} \right) \right)$$

$$y = \tan^{-1} \left(\tan \left(\frac{\pi}{2} - \left(\frac{x}{2} \right) \right) \right)$$

$$y = \frac{\pi}{2} - \frac{x}{2}$$

$$\frac{dy}{dx} = -\frac{1}{2}$$

41.

$$(i) \text{ Given } y = \tan^{-1} \left(\frac{a+x}{1-ax} \right)$$

$$y = \tan^{-1}(x) + \tan^{-1}(a)$$

$$\frac{dy}{dx} = \frac{1}{1+x^2}$$

(ii)

$$y = \tan^{-1} \left\{ \frac{a \cos x - b \sin x}{b \cos x + a \sin x} \right\} - \frac{\pi}{2} < x < \frac{\pi}{2}$$

$$y = \tan^{-1} \left(\frac{\frac{a}{b} - \tan x}{1 + \frac{a}{b} \tan x} \right)$$

$$y = \tan^{-1} \left(\frac{a}{b} \right) - \tan^{-1}(\tan x)$$

$$y = \tan^{-1} \left(\frac{a}{b} \right) - x$$

$$\frac{dy}{dx} = -1$$

(iii)

$$y = \tan^{-1} \left(\frac{3a^2x - x^3}{a^3 - 3ax^2} \right), \quad -\frac{1}{\sqrt{3}} < \frac{x}{a} < \frac{1}{\sqrt{3}}$$

$$y = \tan^{-1} \left(\frac{3 \left(\frac{x}{a} \right) - \left(\frac{x}{a} \right)^3}{1 - 3 \left(\frac{x}{a} \right)^2} \right)$$

$$y = \tan^{-1} \left(\frac{3x}{a} \right)$$

$$\frac{dy}{dx} = \frac{1}{1 + \left(\frac{3x}{a} \right)^2} \times \frac{3}{a}$$

$$\frac{dy}{dx} = \frac{3a}{9x^2 + a^2}$$

(iv)

$$y = \tan^{-1} \left(\sqrt{\frac{a-x}{a+x}} \right)$$

Put

$$x = a \cos \theta$$

$$y = \tan^{-1} \left(\sqrt{\frac{a(1-\cos \theta)}{a(1+\cos \theta)}} \right)$$

$$y = \tan^{-1} \left(\frac{\sin(\theta/2)}{\cos(\theta/2)} \right) = \tan^{-1} \left(\tan \left(\frac{\theta}{2} \right) \right)$$

$$y = \left(\frac{\theta}{2} \right) = \frac{\cos^{-1} x}{2}$$

$$\frac{dy}{dx} = \frac{1}{2\sqrt{1-x^2}}$$

(v)

$$y = \tan^{-1} \left(\frac{\sin x}{1 + \cos x} \right), \quad -\pi < x < \pi$$

$$y = \tan^{-1} \left(\frac{2 \sin(x/2) \cos(x/2)}{2 \cos^2(x/2)} \right)$$

$$y = \tan^{-1} \left(\tan \left(\frac{x}{2} \right) \right) = \frac{x}{2}$$

$$\frac{dy}{dx} = \frac{1}{2}$$

(vi)

$$y = \tan^{-1} \left(\frac{a + b \tan x}{b - a \tan x} \right)$$

$$y = \tan^{-1} \left(\frac{\frac{a}{b} + \tan x}{1 - \frac{a}{b} \tan x} \right)$$

$$y = \tan^{-1} \left(\frac{a}{b} \right) + \tan^{-1}(\tan x)$$

$$y = \tan^{-1} \left(\frac{a}{b} \right) + x$$

$$\frac{dy}{dx} = 1$$

(vii)

$$y = \tan^{-1} \left(\frac{a + bx}{b - ax} \right)$$

$$y = \tan^{-1} \left(\frac{\frac{a}{b} + x}{1 - \frac{a}{b} x} \right) = \tan^{-1} \left(\frac{a}{b} \right) + \tan^{-1}(x)$$

$$y = \tan^{-1} \left(\frac{a}{b} \right) + \tan^{-1}(x)$$

$$\frac{dy}{dx} = \frac{1}{1+x^2}$$

(viii)

$$y = \tan^{-1} \left(\frac{x}{1+6x^2} \right)$$

$$y = \tan^{-1} \left(\frac{3x-2x}{1+6x} \right)$$

$$y = \tan^{-1}(3x) - \tan^{-1}(2x)$$

$$\frac{dy}{dx} = \frac{3}{1+9x^2} - \frac{2}{1+4x^2}$$

(ix) Given

$$y = \sin^{-1} \left(\frac{\sin x + \cos x}{\sqrt{2}} \right)$$

$$= \sin^{-1} \left(\frac{1}{\sqrt{2}} \sin x + \frac{1}{\sqrt{2}} \cos x \right)$$

$$= \sin^{-1}\left(\sin\left(x + \frac{\pi}{4}\right)\right)$$

$$\text{Thus, } y = \left(x + \frac{\pi}{4}\right)$$

$$\Rightarrow \frac{dy}{dx} = 1$$

$$(x) \quad y = \sin^{-1}\left(\frac{1-x^2}{1+x^2}\right) + \sec^{-1}\left(\frac{1+x^2}{1-x^2}\right), x \in R$$

$$y = \sin^{-1}\left(\frac{1-x^2}{1+x^2}\right) + \cos^{-1}\left(\frac{1-x^2}{1+x^2}\right)$$

$$y = \frac{\pi}{2}$$

$$\frac{dy}{dx} = 0$$

$$(xi) \text{ Given } y = \sin^{-1}\left(\sqrt{\frac{x-1}{x+1}}\right) + \cos^{-1}\left(\sqrt{\frac{x-1}{x+1}}\right)$$

$$\Rightarrow y = \frac{\pi}{2}$$

$$\Rightarrow \frac{dy}{dx} = 0$$

$$(xii) \text{ Given } y = \sin^{-1}\left(\frac{3\sin x + 4\cos x}{5}\right)$$

$$\Rightarrow y = \sin^{-1}\left(\frac{3}{5}\sin x + \frac{4}{5}\cos x\right)$$

$$\Rightarrow y = \sin^{-1}(\cos \alpha \sin x + \sin \alpha \cos x)$$

$$\Rightarrow y = \sin^{-1}(\sin(x + \alpha)) = (x + \alpha)$$

$$\Rightarrow \frac{dy}{dx} = 1, \text{ where } \alpha = \tan^{-1}\left(\frac{4}{3}\right)$$

$$(xiii) \text{ Given } y = \tan^{-1}\left(\frac{1-\cos x}{\sin x}\right)$$

$$= \tan^{-1}\left(\frac{2\sin^2\left(\frac{x}{2}\right)}{2\sin\left(\frac{x}{2}\right)\cos\left(\frac{x}{2}\right)}\right)$$

$$= \tan^{-1}\left(\frac{\sin\left(\frac{x}{2}\right)}{\cos\left(\frac{x}{2}\right)}\right)$$

$$= \tan^{-1}\left(\tan\left(\frac{x}{2}\right)\right) = \frac{x}{2}$$

$$\text{Thus, } \frac{dy}{dx} = \frac{1}{2}$$

$$(xiv) \text{ Given } y = \tan^{-1}\left(\frac{\cos x}{1-\sin x}\right)$$

$$= \tan^{-1}\left(\frac{2\cos^2\left(\frac{x}{2}\right)}{2\sin\left(\frac{x}{2}\right)\cos\left(\frac{x}{2}\right)}\right)$$

$$= \tan^{-1}\left(\frac{\cos\left(\frac{x}{2}\right)}{\sin\left(\frac{x}{2}\right)}\right)$$

$$= \tan^{-1}\left(\cot\left(\frac{x}{2}\right)\right)$$

$$= \tan^{-1}\left(\tan\left(\frac{\pi}{2} - \frac{x}{2}\right)\right)$$

$$\Rightarrow y = \left(\frac{\pi}{2} - \frac{x}{2}\right)$$

$$\Rightarrow \frac{dy}{dx} = -\frac{1}{2}$$

$$42. \text{ Given } y = \sin^{-1}\left(\frac{2x}{1+x^2}\right) + \cos^{-1}\left(\frac{1-x^2}{1+x^2}\right)$$

$$y = 2\tan^{-1}x + 2\tan^{-1}x = 4\tan^{-1}x$$

$$\frac{dy}{dx} = \frac{4}{1+x^2}$$

Hence, the result.

$$43. \text{ Given } y = \sin^{-1}\left(\frac{x}{\sqrt{1+x^2}}\right) + \cos^{-1}\left(\frac{1}{\sqrt{x^2+1}}\right)$$

$$\text{Put } x = \tan \theta$$

$$\text{Then } y = \sin^{-1}\left(\frac{\tan \theta}{\sqrt{1+\tan^2 \theta}}\right) + \cos^{-1}\left(\frac{1}{\sqrt{1+\tan^2 \theta}}\right)$$

$$y = \sin^{-1}\left(\frac{\tan \theta}{\sec \theta}\right) + \cos^{-1}\left(\frac{1}{\sec \theta}\right)$$

$$y = \sin^{-1}(\sin \theta) + \cos^{-1}(\cos \theta)$$

$$y = \theta + \theta = 2\theta = 2\tan^{-1}x$$

$$\frac{dy}{dx} = \frac{2}{1+x^2}$$

$$44. \text{ Given } y = \tan^{-1}\left(\frac{\sqrt{1+x^2} + \sqrt{1-x^2}}{\sqrt{1+x^2} - \sqrt{1-x^2}}\right)$$

$$\text{Put } x^2 = \cos(2\theta)$$

$$\text{Thus, } y = \tan^{-1}\left(\frac{\sqrt{1+\cos(2\theta)} + \sqrt{1-\cos(2\theta)}}{\sqrt{1+\cos(2\theta)} - \sqrt{1-\cos(2\theta)}}\right)$$

$$y = \tan^{-1}\left(\frac{\sqrt{2\cos^2 \theta} + \sqrt{2\sin^2 \theta}}{\sqrt{2\cos^2 \theta} - \sqrt{2\sin^2 \theta}}\right)$$

$$y = \tan^{-1}\left(\frac{\cos \theta + \sin \theta}{\cos \theta - \sin \theta}\right)$$

$$y = \tan^{-1}\left(\frac{1 + \tan \theta}{1 - \tan \theta}\right)$$

$$y = \tan^{-1}\left(\tan\left(\frac{\pi}{4} + \theta\right)\right) = \left(\frac{\pi}{4} + \theta\right)$$

$$y = \left(\frac{\pi}{4} + \theta\right) = \frac{\pi}{4} + \frac{1}{2}\cos^{-1}(x^2)$$

$$\frac{dy}{dx} = -\frac{1 \times 2x}{2\sqrt{1-x^4}} = -\frac{x}{\sqrt{1-x^4}}$$

45. Given $y = \cot^{-1} \left\{ \frac{\sqrt{1+\sin x} + \sqrt{1-\sin x}}{\sqrt{1+\sin x} - \sqrt{1-\sin x}} \right\}$

$$= \cot^{-1} \left\{ \frac{\sqrt{\left(\cos\left(\frac{x}{2}\right) + \sin\left(\frac{x}{2}\right)\right)^2} + \sqrt{\left(\cos\left(\frac{x}{2}\right) - \sin\left(\frac{x}{2}\right)\right)^2}}{\sqrt{\left(\cos\left(\frac{x}{2}\right) + \sin\left(\frac{x}{2}\right)\right)^2} - \sqrt{\left(\cos\left(\frac{x}{2}\right) - \sin\left(\frac{x}{2}\right)\right)^2}} \right\}$$

$$= \cot^{-1} \left\{ \frac{\left(\cos\left(\frac{x}{2}\right) + \sin\left(\frac{x}{2}\right)\right) + \left(\cos\left(\frac{x}{2}\right) - \sin\left(\frac{x}{2}\right)\right)}{\left(\cos\left(\frac{x}{2}\right) + \sin\left(\frac{x}{2}\right)\right) - \left(\cos\left(\frac{x}{2}\right) - \sin\left(\frac{x}{2}\right)\right)} \right\}$$

$$= \cot^{-1} \left\{ \frac{2\cos\left(\frac{x}{2}\right)}{2\sin\left(\frac{x}{2}\right)} \right\}$$

$$= \cot^{-1} \left\{ \cot\left(\frac{x}{2}\right) \right\} = \frac{x}{2}$$

$$\frac{dy}{dx} = \frac{1}{2} = \text{independent of } x$$

Hence, the result.

46. Given $y = \tan^{-1} \left(\frac{2x}{1-x^2} \right) + \sec^{-1} \left(\frac{1+x^2}{1-x^2} \right)$

$$y = \tan^{-1} \left(\frac{2x}{1-x^2} \right) + \cos^{-1} \left(\frac{1-x^2}{1+x^2} \right)$$

$$y = 2\tan^{-1} x + 2\tan^{-1} x = 4\tan^{-1} x$$

$$\frac{dy}{dx} = \frac{4}{1+x^2}$$

47. Given $y = \sec^{-1} \left(\frac{x-1}{x+1} \right) + \sin^{-1} \left(\frac{x+1}{x-1} \right)$

$$y = \cos^{-1} \left(\frac{x+1}{x-1} \right) + \sin^{-1} \left(\frac{x+1}{x-1} \right)$$

$$y = \frac{\pi}{2}$$

$$\frac{dy}{dx} = 0$$

48. Given $y = \sin \left[2\tan^{-1} \left\{ \sqrt{\frac{1-x}{1+x}} \right\} \right]$

Put $x = \cos \theta$

$$y = \sin \left[2\tan^{-1} \left\{ \sqrt{\frac{1-\cos \theta}{1+\cos \theta}} \right\} \right]$$

$$y = \sin \left[2\tan^{-1} \left\{ \sqrt{\frac{2\sin^2(\theta/2)}{2\cos^2(\theta/2)}} \right\} \right]$$

$$y = \sin \left[2\tan^{-1} \left(\tan\left(\frac{\theta}{2}\right) \right) \right]$$

$$y = \sin \left[2\left(\frac{\theta}{2}\right) \right] = \sin \theta$$

$$y = \sqrt{1-\cos^2 \theta} = \sqrt{1-x^2}$$

$$\frac{dy}{dx} = \frac{1 \times (-2x)}{2\sqrt{1-x^2}} = -\frac{x}{\sqrt{1-x^2}}$$

49. Given $y = \cos^{-1}(2x) + 2\cos^{-1}(\sqrt{1-4x^2})$

Put $2x = \sin \theta$

Thus, $y = \cos^{-1}(\sin \theta) + 2\cos^{-1}\sqrt{1-\sin^2 \theta}$

$$y = \cos^{-1}(\sin \theta) + 2\cos^{-1}(\cos \theta)$$

$$y = \cos^{-1} \left(\cos\left(\frac{\pi}{2} - \theta\right) \right) + 2\cos^{-1}(\cos \theta)$$

$$y = \left(\frac{\pi}{2} - \theta\right) + 2\theta$$

$$y = \left(\frac{\pi}{2} + \theta\right) = \frac{\pi}{2} + \sin^{-1}(2x)$$

$$\frac{dy}{dx} = \frac{2}{\sqrt{1-4x^2}}$$

50. Given $y = \tan^{-1} \left(\frac{\sqrt{1+x} - \sqrt{1-x}}{\sqrt{1+x} + \sqrt{1-x}} \right)$

Put $x = \cos \theta$

$$y = \tan^{-1} \left(\frac{\sqrt{1+\cos \theta} - \sqrt{1-\cos \theta}}{\sqrt{1+\cos \theta} + \sqrt{1-\cos \theta}} \right)$$

$$y = \tan^{-1} \left(\frac{\cos(\theta/2) - \sin(\theta/2)}{\cos(\theta/2) + \sin(\theta/2)} \right)$$

$$y = \tan^{-1} \left(\frac{1 - \tan\left(\frac{\theta}{2}\right)}{1 + \tan\left(\frac{\theta}{2}\right)} \right)$$

$$y = \tan^{-1} \left(\tan\left(\frac{\pi}{4} - \frac{\theta}{2}\right) \right) = \left(\frac{\pi}{4} - \frac{\theta}{2}\right)$$

$$y = \left(\frac{\pi}{2} - \frac{\theta}{2}\right) = \frac{\pi}{2} - \frac{\cos^{-1} x}{2}$$

$$\frac{dy}{dx} = \frac{1}{2\sqrt{1-x^2}}$$

51. $y = \cos^{-1} x + \cos^{-1} \left(\frac{x}{2} + \frac{\sqrt{3-3x^2}}{2} \right)$

$$y = \cos^{-1} x + \cos^{-1} \left(\frac{1}{2}x + \frac{\sqrt{3}}{2}\sqrt{1-x^2} \right)$$

$$y = \cos^{-1} x + \cos^{-1} \left(\frac{1}{2} \right) - \cos^{-1}(x)$$

$$y = \cos^{-1} \left(\frac{1}{2} \right) = \frac{\pi}{3}$$

$$\frac{dy}{dx} = 0$$

52. We have $y = \sin^{-1}(x\sqrt{1-x} - \sqrt{x-x^3})$

$$\Rightarrow y = \sin^{-1}(x\sqrt{1-x} - \sqrt{x}\sqrt{1-x^2})$$

$$\Rightarrow y = \sin^{-1}x - \sin^{-1}(\sqrt{x})$$

$$\Rightarrow \frac{dy}{dx} = \frac{1}{\sqrt{1-x^2}} - \frac{\sqrt{x}}{2\sqrt{1-x}}$$

53. Given $y = f(x) = x^3 + x^5 + x^7$

$$\Rightarrow \frac{dy}{dx} = 3x^2 + 5x^4 + 7x^6$$

Now, $g'(y) = \frac{dx}{dy} = \frac{1}{3x^2 + 5x^4 + 7x^6}$

When $y = 3$, $3 = x^3 + x^5 + x^7$

Thus, $x = 1$

Therefore, $g'(3) = \frac{1}{3 + 5 + 7} = \frac{1}{15}$

54. Given $y = x^5 + 2x^3 + 2x$

$$\frac{dy}{dx} = 5x^4 + 6x^2 + 2$$

Now, $g'(y) = \frac{dx}{dy} = \frac{1}{5x^4 + 6x^2 + 2}$

when $y = -5$, then $x = -1$

Therefore, $g'(-5) = \frac{1}{5 + 6 + 2} = \frac{1}{13}$

55. Given $y = f(x) = x^3 + 2x^2 + 3x + 4$

$$\Rightarrow \frac{dy}{dx} = 3x^2 + 4x + 3$$

Now, $g'(y) = \frac{dx}{dy} = \frac{1}{3x^2 + 4x + 3}$

When $y = 4$, $x^3 + 2x^2 + 3x + 4 = 4$

Thus, $x = 0$

Therefore, $g'(4) = \frac{1}{0 + 0 + 3} = \frac{1}{3}$

56. Given f and g are inverse of each other

Thus, $f(g(x)) = g(f(x)) = x$

$$\Rightarrow f'(g(x)) \cdot g'(x) = 1$$

$$\Rightarrow g'(x) = \frac{1}{f'(g(x))}$$

$$\Rightarrow g'(x) = \frac{1}{1 + (g(x))^n}$$

57. We have $2x^2 + 3xy + 3y^2 = 1$

$$\Rightarrow 4x + 3x \frac{dy}{dx} + 3y + 6y^2 \frac{dy}{dx} = 0$$

$$\Rightarrow (3x + 6y^2) \frac{dy}{dx} = -(4x + 3y)$$

$$\Rightarrow \frac{dy}{dx} = -\left(\frac{4x + 3y}{3x + 6y^2}\right)$$

58. We have, $e^x + e^y = e^{x+y}$

$$\Rightarrow e^x + e^y \frac{dy}{dx} = e^{x+y} \left(1 + \frac{dy}{dx}\right) = e^{x+y} + e^{x+y} \frac{dy}{dx}$$

$$\Rightarrow e^x + e^y \frac{dy}{dx} = e^{x+y} + e^{x+y} \frac{dy}{dx}$$

$$\Rightarrow (e^y - e^{x+y}) \frac{dy}{dx} = (e^{x+y} - e^x)$$

$$\Rightarrow \frac{dy}{dx} = \left(\frac{e^{x+y} - e^x}{e^y - e^{x+y}}\right) = \left(\frac{e^x + e^y - e^x}{e^y - e^x - e^y}\right)$$

$$\Rightarrow \frac{dy}{dx} = \frac{e^y}{e^x} = -e^{y-x}$$

59. We have $x\sqrt{1+y} + y\sqrt{1+x} = 0$

$$\Rightarrow x\sqrt{1+y} = -y\sqrt{1+x}$$

$$\Rightarrow x^2(1+y) = y^2(1+x)$$

$$\Rightarrow x^2 + x^2y = y^2 + xy^2$$

$$\Rightarrow x^2 - y^2 = xy^2 - x^2y$$

$$\Rightarrow (x+y)(x-y) = -xy(x-y)$$

$$\Rightarrow (x+y) = -xy$$

$$\Rightarrow y + xy = -x$$

$$\Rightarrow y(1+x) = -x$$

$$\Rightarrow y = -\frac{x}{1+x} = -\left(1 - \frac{1}{1+x}\right)$$

$$\Rightarrow y = \left(-1 - \frac{1}{1+x}\right)$$

$$\Rightarrow \frac{dy}{dx} = -\frac{1}{(1+x)^2}$$

60. We have $y\sqrt{1-x^2} + x\sqrt{1-y^2} = 1$... (i)

Put $x = \sin\theta$ and $y = \sin\phi$

(i) reduces to $\sin\phi \cos\theta + \sin\theta \cos\phi = 1$

$$\Rightarrow \sin(\theta + \phi) = 1$$

$$\Rightarrow \theta + \phi = \sin^{-1}(1)$$

$$\Rightarrow \sin^{-1}x + \sin^{-1}y = \sin^{-1}(1)$$

$$\Rightarrow \frac{1}{\sqrt{1-x^2}} + \frac{1}{\sqrt{1-y^2}} \frac{dy}{dx} = 0$$

$$\Rightarrow \frac{1}{\sqrt{1-y^2}} \frac{dy}{dx} = -\frac{1}{\sqrt{1-x^2}}$$

$$\Rightarrow \frac{dy}{dx} = -\frac{\sqrt{1-y^2}}{\sqrt{1-x^2}}$$

$$\Rightarrow \frac{dy}{dx} = -\sqrt{\frac{1-y^2}{1-x^2}}$$

Hence, the result.

61. Given $\sin y = x \sin(a+y)$

$$\Rightarrow x = \frac{\sin y}{\sin(a+y)}$$

Differentiate w.r.t y , we get,

$$\Rightarrow \frac{dx}{dy} = \frac{\sin(a+y) \cdot \cos y - \sin y \cdot \cos(a+y)}{[\sin(a+y)]^2}$$

$$\Rightarrow \frac{dx}{dy} = \frac{\sin(a+y-y)}{[\sin(a+y)]^2}$$

$$\Rightarrow \frac{dx}{dy} = \frac{\sin a}{\sin^2(a+y)}$$

$$\Rightarrow \frac{dx}{dy} = \frac{1}{\frac{dx}{dy}} = \frac{\sin^2(a+y)}{\sin a}$$

62. We have $\log(x+y) = 2xy$

$$\Rightarrow \frac{1}{(x+y)} \left(1 + \frac{dy}{dx}\right) = 2 \left(x \frac{dy}{dx} + y \cdot 1\right) \quad \dots(i)$$

when $x = 0$, then $y = 1$

Put $x = 0$ and $y = 1$ in (i), we get,

$$\Rightarrow \frac{1}{(0+1)} \left(1 + \frac{dy}{dx}\right) = 2 \left(0 \cdot \frac{dy}{dx} + 1 \cdot 1\right)$$

$$\Rightarrow \left(1 + \frac{dy}{dx}\right) = 2$$

$$\Rightarrow \frac{dy}{dx} = 1.$$

63. Given $\sqrt{1-x^6} + \sqrt{1-y^6} = a(x^3 - y^3)$

Put $x^3 = \sin \theta$, $y^3 = \sin \phi$

The given equation reduces to

$$\cos \theta + \cos \phi = a(\sin \theta - \sin \phi)$$

$$\Rightarrow 2 \cos \left(\frac{\theta + \phi}{2}\right) \cos \left(\frac{\theta - \phi}{2}\right)$$

$$= 2a \cos \left(\frac{\theta + \phi}{2}\right) \sin \left(\frac{\theta - \phi}{2}\right)$$

$$\Rightarrow \cos \left(\frac{\theta - \phi}{2}\right) = a \sin \left(\frac{\theta - \phi}{2}\right)$$

$$\Rightarrow \cos \left(\frac{\theta - \phi}{2}\right) = a$$

$$\Rightarrow \left(\frac{\theta - \phi}{2}\right) = \cot^{-1}(a)$$

$$\Rightarrow (\theta - \phi) = 2 \cot^{-1}(a)$$

$$\Rightarrow \sin^{-1}(x^3) - \sin^{-1}(y^3) = 2 \cot^{-1}(a)$$

Differentiating w.r.t x , we get,

$$\Rightarrow \frac{3x^2}{\sqrt{1-x^6}} - \frac{3y^2}{\sqrt{1-y^6}} \frac{dy}{dx} = 0$$

$$\Rightarrow \frac{3y^2}{\sqrt{1-y^6}} \frac{dy}{dx} = \frac{3x^2}{\sqrt{1-x^6}}$$

$$\Rightarrow \frac{dy}{dx} = \frac{x^2 \sqrt{1-y^6}}{y^2 \sqrt{1-x^6}}$$

64. Given, $x^2 + y^2 = t - \frac{1}{t}$ and $x^4 + y^4 = t^2 + \frac{1}{t^2}$

$$\text{Now, } x^2 + y^2 = t - \frac{1}{t}$$

$$\Rightarrow (x^2 + y^2)^2 = \left(t - \frac{1}{t}\right)^2$$

$$\Rightarrow (x^2 + y^2)^2 = t^2 + \frac{1}{t^2} - 2$$

$$\Rightarrow x^4 + y^4 + 2x^2y^2 = t^2 + \frac{1}{t^2} - 2$$

$$\Rightarrow t^2 + \frac{1}{t^2} + 2x^2y^2 = t^2 + \frac{1}{t^2} - 2$$

$$\Rightarrow 2x^2y^2 = -2$$

$$\Rightarrow x^2y^2 = -1$$

$$\Rightarrow y^2 = \frac{-1}{x^2}$$

$$\Rightarrow 2y \frac{dy}{dx} = \frac{2}{x^3}$$

$$\Rightarrow \frac{dy}{dx} = \frac{1}{x^3y}$$

65. Given $\sec \left(\frac{x+y}{x-y}\right) = a$

$$\Rightarrow \left(\frac{x+y}{x-y}\right) = \frac{\sec^{-1}(a)}{1}$$

$$\Rightarrow \left(\frac{x+y+x-y}{x+y-x+y}\right) = \frac{\sec^{-1}(a) + 1}{\sec^{-1}(a) - 1}$$

$$\Rightarrow \frac{x}{y} = \frac{\sec^{-1}(a) + 1}{\sec^{-1}(a) - 1}$$

$$\Rightarrow \frac{y}{x} = \frac{\sec^{-1}(a) - 1}{\sec^{-1}(a) + 1}$$

$$\Rightarrow y = \left(\frac{\sec^{-1}(a) - 1}{\sec^{-1}(a) + 1} \right) x$$

$$\Rightarrow \frac{dy}{dx} = \frac{\sec^{-1}(a) - 1}{\sec^{-1}(a) + 1}$$

$$\Rightarrow \frac{dy}{dx} = \frac{y}{x}$$

Hence, the result.

66. Given $\tan^{-1} \left(\frac{x^2 - y^2}{x^2 + y^2} \right) = a$

$$\Rightarrow \left(\frac{x^2 - y^2}{x^2 + y^2} \right) = \frac{\tan a}{1}$$

$$\Rightarrow \left(\frac{x^2 - y^2 + x^2 + y^2}{x^2 - y^2 - x^2 - y^2} \right) = \frac{\tan a + 1}{\tan a - 1}$$

$$\Rightarrow \frac{-x^2}{y^2} = \frac{\tan a + 1}{\tan a - 1}$$

$$\Rightarrow \frac{y^2}{x^2} = \frac{1 - \tan a}{\tan a + 1}$$

$$\Rightarrow y^2 = \left(\frac{1 - \tan a}{\tan a + 1} \right) x^2$$

$$\Rightarrow 2y \frac{dy}{dx} = 2 \left(\frac{1 - \tan a}{\tan a + 1} \right) x$$

$$\Rightarrow \frac{dy}{dx} = \frac{x}{y} \left(\frac{1 - \tan a}{1 + \tan a} \right)$$

Hence, the result.

67. Given $xy = 1$

$$\Rightarrow y = \frac{1}{x}$$

$$\Rightarrow \frac{dy}{dx} = -\frac{1}{x^2}$$

$$\Rightarrow \frac{dy}{dx} = -y^2$$

$$\Rightarrow \frac{dy}{dx} + y^2 = 0$$

Hence, the result.

68. Given $y = x \sin y$

$$\Rightarrow \frac{dy}{dx} = \sin y + x \cos y \frac{dy}{dx}$$

$$\Rightarrow \frac{dy}{dx} (1 - x \cos y) = \sin y$$

$$\Rightarrow \frac{dy}{dx} = \frac{\sin y}{(1 - x \cos y)}$$

$$\Rightarrow \frac{dy}{dx} = \frac{y}{x(1 - x \cos y)}$$

Hence, the result.

69. Given $\cos y = x \cos(a + y)$

$$\Rightarrow x = \frac{\cos y}{\cos(a + y)}$$

$$\Rightarrow \frac{dx}{dy} = \frac{-\cos(a + y) \sin y + \cos y \sin(a + y)}{\cos^2(a + y)}$$

$$\Rightarrow \frac{dx}{dy} = \frac{\sin(a + y - y)}{\cos^2(a + y)}$$

$$\Rightarrow \frac{dx}{dy} = \frac{\sin(a)}{\cos^2(a + y)}$$

$$\Rightarrow \frac{dx}{dy} = \frac{\cos^2(a + y)}{\sin(a)}$$

Hence, the result.

70. Given $y = x^{\sin x}$

$$\Rightarrow \log(y) = \log(x^{\sin x})$$

$$\Rightarrow \log(y) = \sin x \log x$$

$$\Rightarrow \frac{1}{y} \frac{dy}{dx} = \frac{\sin x}{x} + \cos x \log x$$

$$\Rightarrow \frac{dy}{dx} = y \left(\frac{\sin x}{x} + \cos x \log x \right)$$

$$\Rightarrow \frac{dy}{dx} = x^{\sin x} \left(\frac{\sin x}{x} + \cos x \log x \right)$$

71. Given $y = (\sin x)^{\cos x}$

$$\Rightarrow y = e^{\log(\sin x)^{\cos x}}$$

$$\Rightarrow y = e^{\cos x \log(\sin x)}$$

$$\Rightarrow \frac{dy}{dx} = e^{\cos x \log(\sin x)} \left(\cos x \cdot \frac{\cos x}{\sin x} - \sin x \log \sin x \right)$$

$$\Rightarrow \frac{dy}{dx} = (\sin x)^{\cos x} (\cos x \cdot \cot x - \sin x \log \sin x)$$

72. Given $y = (\tan x)^{\cot x} + (\cot x)^{\tan x}$

$$\Rightarrow y = e^{\log(\tan x)^{\cot x}} + e^{\log(\cot x)^{\tan x}}$$

$$\Rightarrow y = e^{\cot x \log(\tan x)} + e^{\tan x \log(\cot x)}$$

$$\Rightarrow \frac{dy}{dx} = e^{\cot x \log(\tan x)} \left(\cot x \cdot \frac{\sec^2 x}{\tan x} - \operatorname{cosec}^2 x \cdot \log(\tan x) \right)$$

$$+ e^{\tan x \log(\cot x)} \left(\tan x \cdot \frac{-\operatorname{cosec}^2 x}{\cot x} + \sec^2 x \cdot \log(\cot x) \right)$$

$$\Rightarrow \frac{dy}{dx} = (\tan x)^{\cot x} (\operatorname{cosec}^2 x - \operatorname{cosec}^2 x \cdot \log(\tan x)) \\ + (\cot x)^{\tan x} (-\sec^2 x + \sec^2 x \cdot \log(\cot x))$$

73. Given $x^m x^n = (x + y)^{m+n}$

$$\Rightarrow \log(x^m x^n) = \log((x + y)^{m+n})$$

$$\Rightarrow m \log x + n \log y = (m + n) \log(x + y)$$

$$\Rightarrow \frac{m}{x} + \frac{n}{y} \frac{dy}{dx} = \frac{(m + n)}{(x + y)} \left(1 + \frac{dy}{dx} \right)$$

$$\Rightarrow \left(\frac{n}{y} - \frac{m + n}{x + y} \right) \frac{dy}{dx} = \left(\frac{m + n}{x + y} - \frac{m}{x} \right)$$

$$\Rightarrow \left(\frac{nx + ny - my - ny}{y(x + y)} \right) \frac{dy}{dx}$$

$$= \left(\frac{mx + nx - mx - my}{x(x + y)} \right)$$

$$\Rightarrow \left(\frac{nx - my}{y(x + y)} \right) \frac{dy}{dx} = \left(\frac{nx - my}{x(x + y)} \right)$$

$$\Rightarrow \left(\frac{1}{y} \right) \frac{dy}{dx} = \left(\frac{1}{x} \right)$$

$$\Rightarrow \frac{dy}{dx} = \frac{y}{x}$$

74. Given $y = \left(1 + \frac{1}{x}\right)^x + x^{\left(1 + \frac{1}{x}\right)}$

$$\Rightarrow y = e^{x \log\left(1 + \frac{1}{x}\right)} + e^{\left(1 + \frac{1}{x}\right) \log x}$$

$$\Rightarrow \frac{dy}{dx} = e^{x \log\left(1 + \frac{1}{x}\right)} \left(\log\left(x + \frac{1}{x}\right) + x \cdot \frac{1}{\left(1 + \frac{1}{x}\right)} \cdot -\frac{1}{x^2} \right)$$

$$+ e^{\left(1 + \frac{1}{x}\right) \log x} \left(\left(1 + \frac{1}{x}\right) \frac{1}{x} - \frac{1}{x^2} \log x \right)$$

$$\Rightarrow \frac{dy}{dx} = \left(1 + \frac{1}{x}\right)^x \left(\log\left(x + \frac{1}{x}\right) - \frac{1}{(1 + x)} \right)$$

$$+ x^{\left(1 + \frac{1}{x}\right)} \left(\left(1 + \frac{1}{x}\right) \frac{1}{x} - \frac{1}{x^2} \log x \right)$$

$$\Rightarrow \left(\frac{dy}{dx} \right)_{x=1} = 2 \left(\log 2 - \frac{1}{2} \right) + 2 = \log 4 + 1$$

75. Given $y = \sin x \cdot \sin 2x \cdot \sin 3x \dots \sin(2014)x$

$$\log y = \log(\sin x \cdot \sin 2x \dots \sin(2014)x)$$

$$\log y = \log \sin x + \log \sin 2x + \dots + \log \sin(2014)x$$

$$\frac{1}{y} \frac{dy}{dx} = \cot x + 2 \cot 2x + \dots + 2014 \cot(2014)x$$

$$\frac{dy}{dx} = y(\cot x + 2 \cot 2x + \dots + 2014 \cot(2014)x)$$

76. Given $x^y = e^{x-y}$

$$\Rightarrow y \log x = (x - y)$$

$$\Rightarrow y(1 + \log x) = x$$

$$\Rightarrow y = \frac{x}{(1 + \log x)}$$

$$\Rightarrow \frac{dy}{dx} = \frac{(1 + \log x) - x \cdot \frac{1}{x}}{(1 + \log x)^2}$$

$$\Rightarrow \frac{dy}{dx} = \frac{(1 + \log x) - 1}{(1 + \log x)^2}$$

$$\Rightarrow \frac{dy}{dx} = \frac{\log x}{(1 + \log x)^2}$$

Hence, the result.

77. Given $x^y = y^x$

$$\Rightarrow y \log x = x \log y$$

$$\Rightarrow \frac{y}{x} + \log x \frac{dy}{dx} = \log y + \frac{x}{y} \frac{dy}{dx}$$

$$\Rightarrow \left(\log x - \frac{x}{y} \right) \frac{dy}{dx} = \left(\log y - \frac{y}{x} \right)$$

$$\Rightarrow \frac{dy}{dx} = \frac{\left(\log y - \frac{y}{x} \right)}{\left(\log x - \frac{x}{y} \right)} = \frac{y}{x} \left(\frac{x \log y - y}{y \log x - x} \right)$$

78. Given $e^y = y^x$,

$$\Rightarrow \log(e^y) = \log(y^x)$$

$$\Rightarrow y \log(e) = x \log(y)$$

$$\Rightarrow y = x \log(y)$$

$$\Rightarrow \frac{dy}{dx} = \log(y) + \frac{x}{y} \frac{dx}{dy}$$

$$\Rightarrow \frac{dy}{dx} \left(1 - \frac{x}{y} \right) = \log(y)$$

$$\Rightarrow \frac{dy}{dx} \left(1 - \frac{1}{\log(y)} \right) = \log(y)$$

$$\Rightarrow \frac{dy}{dx} (\log(y) - 1) = \{\log(y)\}^2$$

$$\Rightarrow \frac{dy}{dx} = \frac{\{\log(y)\}^2}{(\log(y) - 1)}$$

79. Given $x^m y^n = 1$

$$\Rightarrow \log(x^m y^n) = \log(1) = 0$$

$$\Rightarrow m \log x + n \log y = 0$$

$$\Rightarrow \frac{m}{x} + \frac{n}{y} \frac{dy}{dx} = 0$$

$$\Rightarrow \frac{dy}{dx} = -\frac{my}{nx}$$

Hence, the result.

80. Given $e^{x+y} = x$

$$\Rightarrow (x + y) = \log x$$

$$\Rightarrow 1 + \frac{dy}{dx} = \frac{1}{x}$$

$$\Rightarrow \frac{dy}{dx} = \frac{1}{x} - 1 = \frac{1-x}{x}$$

82. Given $y = (x+1)(x+2)(x+3)\dots(x+2012)$

$$\log y = \log(x+1) + \log(x+2) + \dots + \log(x+2012)$$

Differentiating w.r.t x , we get,

$$\frac{1}{y} \frac{dy}{dx} = \frac{1}{(x+1)} + \frac{1}{(x+2)} + \dots + \frac{1}{(x+2012)}$$

$$\frac{dy}{dx} = y \left(\frac{1}{(x+1)} + \frac{1}{(x+2)} + \dots + \frac{1}{(x+2012)} \right)$$

Put $x = 0$, we get,

$$\frac{dy}{dx} = (2012)! \left(1 + \frac{1}{2} + \frac{1}{3} + \dots + \frac{1}{2012} \right)$$

83. Given $y = (1+x)(1+x^2)(1+x^4)(1+x^8) \dots$

$$(1+x^{1006})$$

$$\log y = \log(1+x) + \log(1+x^2) + \log(1+x^4) +$$

$$\log(1+x^8) + \dots + \log(1+x^{1006})$$

Differentiating w.r.t x , we get,

$$\frac{1}{y} \frac{dy}{dx} = \frac{1}{(1+x)} + \frac{2x}{1+x^2} + \frac{4x^3}{1+x^4} +$$

$$+ \frac{8x^7}{1+x^8} \dots + \frac{1006x^{1005}}{1+x^{1006}}$$

$$\frac{dy}{dx} = y \left(\frac{1}{1+x} + \frac{2x}{1+x^2} + \dots + \frac{1006x^{1005}}{1+x^{1006}} \right)$$

$$\left(\frac{dy}{dx} \right)_{x=0} = 1(1+1+1+\dots+1)(1006\text{-times}) = 1006.$$

84. We have

$$y = \sqrt{\cos x} + \sqrt{\cos x} + \sqrt{\cos x} + \dots \text{ to } \infty$$

$$\Rightarrow \frac{dy}{dx} = \frac{f'(x)}{(2y-1)}$$

$$\Rightarrow \frac{dy}{dx} = \frac{-\sin x}{2y-1}$$

85. We have $y = \frac{\sin x \cos x \sin x \cos x}{1+1+1+1} \dots$

$$\Rightarrow y = \frac{\sin x}{1 + \frac{\cos x}{1 + \frac{\sin x}{1 + \dots}}} = \frac{\sin x}{1 + \frac{\cos x}{1+y}}$$

$$\Rightarrow y = \frac{(1+y)\cos x}{1+y+\cos x}$$

$$\Rightarrow y + y^2 + y \cos x = \sin x + y \sin x$$

$$\Rightarrow \frac{dy}{dx} + 2y \frac{dy}{dx} + \cos x \frac{dy}{dx} - y \sin x$$

$$= \cos x + \sin x \frac{dy}{dx}$$

$$\Rightarrow (1+2y+\cos x-\sin x) \frac{dy}{dx} = \cos x + y \sin x$$

$$\Rightarrow \frac{dy}{dx} = \frac{\cos x + y \sin x}{(1+2y+\cos x-\sin x)}$$

86. Given $y = \sqrt{x} + \sqrt{x} + \sqrt{x} + \dots \text{ to } \infty$

$$\Rightarrow y = \sqrt{x+y}$$

$$\Rightarrow y^2 = x+y$$

$$\Rightarrow 2y \frac{dy}{dx} = 1 + \frac{dy}{dx}$$

$$\Rightarrow (2y-1) \frac{dy}{dx} = 1$$

$$\Rightarrow \frac{dy}{dx} = \frac{1}{(2y-1)}$$

Hence, the result.

87. Given $y = x^{x \dots \text{to } \infty}$

$$\Rightarrow y = x^y$$

$$\Rightarrow \log(y) = y \log x$$

$$\Rightarrow \frac{1}{y} \frac{dy}{dx} = \frac{y}{x} + \log x \frac{dy}{dx}$$

$$\Rightarrow \left(\frac{1}{y} - \log(x) \right) \frac{dy}{dx} = \frac{y}{x}$$

$$\Rightarrow (1 - y \log(x)) \frac{dy}{dx} = \frac{y^2}{x}$$

$$\Rightarrow \frac{dy}{dx} = \frac{y^2}{x(1 - y \log(x))}$$

Hence, the result.

88. Given $y = \sqrt{\sin x} + \sqrt{\sin x} + \sqrt{\sin x} \dots \text{to } \infty$

$$\Rightarrow y = \sqrt{\sin x + y}$$

$$\Rightarrow y^2 = \sin x + y$$

$$\Rightarrow 2y \frac{dy}{dx} = \cos x + \frac{dy}{dx}$$

$$\Rightarrow (2y - 1) \frac{dy}{dx} = \cos x$$

$$\Rightarrow \frac{dy}{dx} = \frac{\cos x}{(2y - 1)}$$

Hence, the result.

89. Given $y = e^{x+e^{x+e^{x+\dots \text{to } \infty}}}$

$$\Rightarrow y = e^{x+y}$$

$$\Rightarrow \frac{dy}{dx} = e^{x+y} \left(1 + \frac{dy}{dx} \right)$$

$$\Rightarrow \frac{dy}{dx} = y \left(1 + \frac{dy}{dx} \right)$$

$$\Rightarrow (1 - y) \frac{dy}{dx} = y$$

$$\Rightarrow \frac{dy}{dx} = \frac{y}{1 - y}$$

90. Given $y = (\tan x)^{(\tan x)^{(\tan x) \dots \text{to } \infty}}$

$$\Rightarrow y = (\tan x)^y$$

$$\Rightarrow \log(y) = y \log(\tan x)$$

$$\Rightarrow \frac{1}{y} \frac{dy}{dx} = \log(\tan x) \frac{dy}{dx} + \left(\frac{\sec^2 x}{\tan x} \right)^y$$

when $x = \frac{\pi}{4}$, $y = 1$, then

$$\Rightarrow \frac{dy}{dx} = 0 \cdot \frac{dy}{dx} + 2$$

$$\Rightarrow \frac{dy}{dx} = 2$$

91. Given $y = x + \frac{1}{x} + \frac{1}{x} + \frac{1}{x} \dots \text{to } \infty$

$$\Rightarrow y = x + \frac{1}{y}$$

$$\Rightarrow y^2 = xy + 1$$

Differentiating w.r.t x , we get,

$$\Rightarrow 2y \frac{dy}{dx} = y + x \frac{dy}{dx}$$

$$\Rightarrow (2y - x) \frac{dy}{dx} = y$$

$$\Rightarrow \frac{dy}{dx} = \frac{y}{(2y - x)}$$

Hence, the result.

92. Given $y = \frac{\sin x}{1 + \frac{\cos x}{1 + \frac{\sin x}{1 + \frac{\cos x}{\dots \text{to } \infty}}}}$

$$\Rightarrow y = \frac{\sin x}{1 + \frac{\cos x}{1 + y}}$$

$$\Rightarrow y = \frac{(1 + y) \sin x}{(1 + y) + \cos x}$$

$$\Rightarrow y \{(1 + y) + \cos x\} = (1 + y) \sin x$$

$$\Rightarrow y' \{(1 + y) + \cos x\} + y(y' - \sin x) = (1 + y) \cos x + y' \sin x$$

when $x = 0$ and $y = 0$, then

$$\Rightarrow y'(1 + 1) + 0 = 1 + 0$$

$$\Rightarrow 2y' = 1$$

$$\Rightarrow y' = \frac{1}{2}$$

$$\Rightarrow \frac{dy}{dx} = \frac{1}{2}$$

93. Given $y = \frac{x}{1 + \frac{x}{2 + \frac{x}{1 + \frac{x}{2 + \dots \text{to } \infty}}}}$

$$\Rightarrow y = \frac{x}{1 + \frac{x}{2 + y}} = \frac{x(2 + y)}{2 + y + x}$$

$$\Rightarrow y(2 + y + x) = x(2 + y)$$

$$\Rightarrow 2y + y^2 + xy = 2x + xy$$

$$\Rightarrow 2y + y^2 = 2x$$

$$\Rightarrow 2 \frac{dy}{dx} + 2y \frac{dy}{dx} = 2$$

$$\Rightarrow \frac{dy}{dx} + y \frac{dy}{dx} = 1$$

$$\Rightarrow (1 + y) \frac{dy}{dx} = 1$$

$$\Rightarrow \frac{du}{dv} = \frac{du/dx}{dv/dx} = -\frac{1}{2}$$

$$110. \text{ Let } u = \sin^{-1}\left(\frac{2x}{1+x^2}\right) \text{ and } v = \cos^{-1}\left(\frac{1-x^2}{1+x^2}\right)$$

$$\Rightarrow u = 2\tan^{-1}(x) \text{ and } v = 2\tan^{-1}(x)$$

$$\Rightarrow \frac{du}{dv} = \frac{du/dx}{dv/dx} = \frac{2/(1+x^2)}{2/(1+x^2)} = 1$$

$$111. \text{ Let } u = (\sin x), v = f(\cos x)$$

$$\Rightarrow \frac{du}{dv} = \frac{du/dx}{dv/dx}$$

$$\Rightarrow \frac{du}{dv} = \frac{du/dx}{du/dx} = -\frac{f'(\sin x) \cos x}{f'(\cos x) \sin x}$$

$$112. \text{ We have } y = c_1 e^x + c_2 e^{-x}$$

$$\Rightarrow \frac{dy}{dx} = c_1 e^x - c_2 e^{-x}$$

$$\Rightarrow \frac{d^2 y}{dx^2} = c_1 e^x + c_2 e^{-x} = y$$

$$\Rightarrow \frac{d^2 y}{dx^2} - y = 0.$$

$$113. \text{ We have } x = at^2, y = 2 \text{ at}$$

$$\Rightarrow \frac{dx}{dt} = 2at \text{ \& } \frac{dy}{dt} = 2a$$

$$\text{Now, } \frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}} = \frac{2a}{2at} = \frac{1}{t}$$

$$\Rightarrow \frac{d^2 y}{dx^2} = \frac{d}{dx} \left(\frac{dy}{dx} \right)$$

$$\Rightarrow = \frac{d}{dt} \left(\frac{dy}{dx} \right) \cdot \frac{dt}{dx}$$

$$\Rightarrow = \frac{d}{dt} \left(\frac{1}{t} \right) \cdot \frac{dt}{dx}$$

$$\Rightarrow = -\frac{1}{t^2} \cdot \frac{1}{2at}$$

$$\Rightarrow = -\frac{1}{2at^3}$$

$$114. \text{ We have } x = a \cos \theta, y = b \sin \theta$$

$$\Rightarrow \frac{dx}{d\theta} = -a \sin \theta \text{ \& } \frac{dy}{d\theta} = b \cos \theta$$

$$\Rightarrow \frac{dx}{d\theta} = \frac{dy}{dx} = \frac{b \cos \theta}{-a \sin \theta} = -\frac{b}{a} \cot \theta$$

$$\Rightarrow \frac{d^2 y}{dx^2} = \frac{d}{dx} \left(\frac{dy}{dx} \right) = \frac{d}{dx} \left(-\frac{b}{a} \cot \theta \right)$$

$$\Rightarrow \frac{d^2 y}{dx^2} = \frac{d}{d\theta} \left(-\frac{b}{a} \cot \theta \right) \cdot \frac{d\theta}{dx}$$

$$\Rightarrow \frac{d^2 y}{dx^2} = \left(\frac{b}{a} \operatorname{cosec}^2 \theta \right) \cdot \frac{-1}{a \sin \theta}$$

$$= \left(-\frac{b}{a^2} \operatorname{cosec}^3 \theta \right)$$

$$= \left(-\frac{b}{a} \cdot \frac{b^3}{y^3} \right)$$

$$= -\frac{b^4}{ay^3}$$

$$115. \frac{d^2 x}{dy^2} = \frac{d}{dy} \left(\frac{dx}{dy} \right) = \frac{d}{dx} \left(\frac{dx}{dy} \right) \cdot \frac{dx}{dy}$$

$$= \frac{d}{dx} \left(\frac{1}{\frac{dy}{dx}} \right) \cdot \frac{dx}{dy}$$

$$= \frac{-1}{\left(\frac{dy}{dx} \right)^2} \cdot \frac{d^2 y}{dx^2} \cdot \frac{1}{\frac{dy}{dx}}$$

$$= -\left(\frac{dy}{dx} \right)^{-3} \cdot \left(\frac{d^2 y}{dx^2} \right)$$

$$116. \text{ Given } y = e^{2x}$$

$$\Rightarrow \frac{dy}{dx} = 2e^{2x}$$

$$\Rightarrow \frac{d^2 y}{dx^2} = 4e^{2x}$$

$$\Rightarrow \frac{d^2 x}{d^2 y} = \frac{d}{dy} \left(\frac{dx}{dy} \right) \cdot \frac{dx}{dy} = \frac{d}{dx} \left(\frac{dx}{dy} \right) \cdot \frac{dx}{dy}$$

$$= \frac{d}{dx} \left(\frac{1}{\frac{dy}{dx}} \right) \cdot \frac{dx}{dy}$$

$$= \frac{-1}{\left(\frac{dy}{dx} \right)^2} \cdot \frac{d^2 y}{dx^2} \cdot \frac{1}{\frac{dy}{dx}}$$

$$= -\left(\frac{dy}{dx} \right)^{-3} \cdot \left(\frac{d^2 y}{dx^2} \right)$$

$$\text{Thus, } \left(\frac{d^2 y}{dx^2} \right) \left(\frac{d^2 x}{dy^2} \right)$$

$$= \frac{d^2 y}{dx^2} \cdot -\left(\frac{dy}{dx} \right)^{-3} \cdot \frac{d^2 y}{dx^2}$$

$$= \left(\frac{d^2 y}{dx^2} \right)^2 \left(\frac{dy}{dx} \right)^{-3}$$

$$= (4e^{2x})^2 (2e^{2x})^{-3}$$

$$= \frac{16e^{4x}}{8e^{6x}}$$

$$= \frac{2}{e^{2x}}$$

$$= \frac{2}{y}$$

117. We have $(a + bx)e^{\frac{y}{x}} = x$

$$\Rightarrow e^{\frac{y}{x}} = \frac{x}{a + bx}$$

$$\Rightarrow \frac{y}{x} = \log\left(\frac{x}{a + bx}\right)$$

$$\Rightarrow \frac{y}{x} = \log x - \log(a + bx)$$

$$\Rightarrow x \cdot \frac{dy}{dx} - y = \frac{1}{x} - \frac{b}{a + bx} = \frac{a + bx - bx}{x(a + bx)}$$

$$\Rightarrow x \cdot \frac{dy}{dx} - y = \frac{ax^2}{x(a + bx)} = \frac{ax}{a + bx}$$

$$\Rightarrow x \cdot \frac{d^2y}{dx^2} + \frac{dy}{dx} \cdot 1 - \frac{dy}{dx} = \frac{(a + bx) \cdot a - ax \cdot b}{(a + bx)^2}$$

$$\Rightarrow x \cdot \frac{d^2y}{dx^2} = \frac{a^2}{(a + bx)^2}$$

$$\Rightarrow x^3 \frac{d^2y}{dx^2} = \frac{a^2 x^2}{(a + bx)^2} = \left(\frac{ax}{a + bx}\right)^2$$

$$\Rightarrow x^3 \frac{d^2y}{dx^2} = \left(x \frac{dy}{dx} - y\right)^2$$

Hence, the result.

118. We have $x^2 + y^2 = a^2$

$$\Rightarrow \frac{d^2y}{dx^2} = -\frac{b}{a^2} \cot^3 \theta = -\frac{b \left(\frac{b}{y}\right)^3}{a^2 y^3} = \frac{a^2 b^4}{a^2 y^3}$$

$$\Rightarrow 2x + 2y \frac{dy}{dx} = 0$$

$$\Rightarrow x + y \frac{dy}{dx} = 0$$

$$\Rightarrow 1 + \left(\frac{dy}{dx}\right)^2 + y \frac{d^2y}{dx^2} = 0$$

$$\Rightarrow -y \frac{d^2y}{dx^2} = 1 + \left(\frac{dy}{dx}\right)^2$$

$$\Rightarrow \left|\frac{d^2y}{dx^2}\right| = \left|\frac{1 + \left(\frac{dy}{dx}\right)^2}{-y}\right| = \frac{1 + \left(\frac{dy}{dx}\right)^2}{|y|}$$

$$\Rightarrow |y''| = \frac{(1 + y'^2)}{|y|} = \frac{(1 + y'^2)^{3/2}}{a}$$

$$\Rightarrow \frac{1}{a} = \frac{|y''|}{(1 + y'^2)^{3/2}}$$

$$\Rightarrow k = \frac{|y''|}{(1 + y'^2)^{3/2}}$$

$$\left[\because x + yy' = 0 \Rightarrow y' = -\frac{x}{y} \Rightarrow y^2 = \frac{x^2}{y^2} \right]$$

$$\Rightarrow 1 + y'^2 = 1 + \frac{x^2}{y^2} = \frac{x^2 + y^2}{y^2} = \frac{a^2}{y^2}$$

$$\Rightarrow y^2 = \frac{a^2}{1 + y'^2} \Rightarrow |y| = \frac{a}{\sqrt{1 + y'^2}}$$

119. We have $y = e^{ax} \sin bx$

$$\Rightarrow \frac{dy}{dx} = a e^{ax} \sin bx + e^{ax} \cdot b \cos bx$$

$$\Rightarrow \frac{d^2y}{dx^2} = a^2 e^{ax} \sin bx + a b e^{ax} \cos bx + a b e^{ax} \cos bx - b^2 e^{ax} \sin bx$$

$$y^2 - 2ay_1 + (a^2 + b^2)y$$

$$= a^2 e^{ax} \sin bx + a b e^{ax} \cos bx$$

$$+ a b e^{ax} \cos bx - b^2 e^{ax} \sin bx$$

$$- 2a(ae^{ax} \sin bx + be^{ax} \cos bx)$$

$$+ (a^2 + b^2)e^{ax} \sin bx$$

$$= e^{ax}(a^2 \sin bx + 2ab \cos bx - b^2 \sin bx$$

$$- 2a^2 \sin bx - 2ab \cos bx + (a^2 + b^2) \sin bx)$$

$$= 0.$$

Hence, the result.

120. Given $y = 2 \sin x + 3 \cos x$

$$\Rightarrow \frac{dy}{dx} = 2 \cos x - 3 \sin x$$

$$\Rightarrow \frac{d^2y}{dx^2} = -2 \sin x - 3 \cos x$$

$$\Rightarrow \frac{d^2y}{dx^2} = -2(\sin x + 3 \cos x) = -y$$

$$\Rightarrow \frac{d^2y}{dx^2} + y = 0$$

Hence, the result.

121. Given $y = x + \tan x$

$$\Rightarrow \frac{dy}{dx} = 1 + \sec^2 x$$

$$\Rightarrow \frac{d^2y}{dx^2} = 2 \sec^2 x \tan x$$

$$\Rightarrow \cos^2 x \frac{d^2y}{dx^2} = 2 \tan x$$

Now, $\cos^2 x \frac{d^2y}{dx^2} - 2y + 2x$

$$= 2 \tan x - 2(x + \tan x) + 2x$$

$$= 2 \tan x - 2x - 2 \tan x + 2x \\ = 0.$$

122. Given $y = \tan x + \sec x$

$$\Rightarrow \frac{dy}{dx} = \sec^2 x + \sec x \tan x \\ \Rightarrow \frac{dy}{dx} = \frac{1 + \sin x}{\cos^2 x} \\ \Rightarrow \frac{dy}{dx} = \frac{1 + \sin x}{1 - \sin^2 x} = \frac{1}{1 - \sin x} \\ \Rightarrow \frac{dy}{dx} = \frac{-\cos x}{(1 - \sin x)^2} = \frac{\cos x}{(1 - \sin x)^2}$$

Hence, the result.

123. Given $y = A \cos(\log x) + B \sin(\log x)$

$$\Rightarrow \frac{dy}{dx} = -\frac{A \sin(\log x)}{x} + \frac{B \cos(\log x)}{x} \\ \Rightarrow x \frac{dy}{dx} = -A \sin(\log x) + B \cos(\log x) \\ \Rightarrow x \frac{d^2y}{dx^2} + \frac{dy}{dx} = -\frac{A \cos(\log x)}{x} - \frac{B \sin(\log x)}{x} \\ \Rightarrow x^2 \frac{d^2y}{dx^2} + x \frac{dy}{dx} = -(A \cos(\log x) + B \sin(\log x)) \\ \Rightarrow x^2 \frac{d^2y}{dx^2} + x \frac{dy}{dx} = -y \\ \Rightarrow x^2 \frac{d^2y}{dx^2} + x \frac{dy}{dx} + y = 0$$

Hence, the result.

124. Given $y = \tan^{-1} x$

$$\Rightarrow \frac{dy}{dx} = \frac{1}{1 + x^2} \\ \Rightarrow (1 + x^2) \frac{dy}{dx} = 1 \\ \Rightarrow (1 + x^2) \frac{d^2y}{dx^2} + 2x \frac{dy}{dx} = 0 \\ \Rightarrow (1 + x^2) y_2 + 2xy_1 = 0$$

125. Given $y = e^x (\sin x + \cos x)$,

$$\Rightarrow \frac{dy}{dx} = e^x (\sin x + \cos x) + e^x (\cos x - \sin x) \\ \Rightarrow \frac{dy}{dx} = 2e^x \cos x \\ \Rightarrow \frac{d^2y}{dx^2} = 2e^x (\cos x - \sin x)$$

$$\text{Now, } \frac{d^2y}{dx^2} - 2 \frac{dy}{dx} + 2y$$

$$= 2e^x (\cos x - \sin x) - 4e^x \cos x + 2e^x (\sin x + \cos x) \\ = 2e^x (\cos x - \sin x - 2 \cos x + \sin x + \cos x) \\ = 2e^x (0) \\ = 0.$$

Hence, the result.

126. Given $y = \sin^{-1}(x)$

$$\Rightarrow \frac{dy}{dx} = \frac{1}{\sqrt{1 - x^2}} \\ \Rightarrow \frac{d^2y}{dx^2} = \frac{(2x)}{2(1 - x^2)\sqrt{1 - x^2}} \\ \Rightarrow (1 - x^2) \frac{d^2y}{dx^2} = \frac{x}{\sqrt{1 - x^2}} = x \frac{dy}{dx} \\ \Rightarrow (1 - x^2) \frac{d^2y}{dx^2} - x \frac{dy}{dx} = 0$$

Hence, the result.

127.

$$(i) \frac{dy}{dx} = \frac{dy/dt}{dx/dt} = \frac{2a}{2at} = \frac{1}{t} \\ \Rightarrow \frac{d^2y}{dx^2} = \frac{d}{dx} \left(\frac{dy}{dx} \right) = \frac{d}{dx} \left(\frac{1}{t} \right) \\ \Rightarrow \frac{d^2y}{dx^2} = \frac{d}{dt} \left(\frac{1}{t} \right) \frac{dt}{dx} \\ \Rightarrow \frac{d^2y}{dx^2} = -\frac{1}{t^2} \times \frac{1}{2at} = -\frac{1}{2at^3} \\ (ii) \frac{dy}{dx} = \frac{dy/d\theta}{dx/d\theta} = -\frac{3a \sin^2 \theta \cos \theta}{3a \cos^2 \theta \sin \theta} \\ \Rightarrow \frac{dy}{dx} = -\tan \theta \\ \Rightarrow \frac{d^2y}{dx^2} = \frac{d}{dx} \left(\frac{dy}{dx} \right) = \frac{d}{dx} (-\tan \theta) \\ \Rightarrow \frac{d^2y}{dx^2} = \frac{d}{d\theta} (-\tan \theta) \frac{d\theta}{dx} \\ \Rightarrow \frac{d^2y}{dx^2} = \sec^2 \theta \times \frac{1}{3a \cos^2 \theta \sin \theta} \\ = \frac{a^3}{3x(axy)^{1/3}} = \frac{a^{8/3}}{3(x^4 y)^{1/3}} = \frac{1}{3} \left(\frac{a^8}{x^4 y} \right)^{1/3}$$

128. Given $x = a \sec \theta$, $y = b \tan \theta$

$$\Rightarrow \frac{d^2y}{dx^2} = -\frac{b}{a} \operatorname{cosec} \theta \cot \theta \times \frac{1}{a \sec \theta \tan \theta} \\ \Rightarrow \frac{d^2y}{dx^2} = -\frac{b}{a^2} \cot^3 \theta = -\frac{b}{a^2} \left(\frac{b}{y} \right)^3 = -\frac{b^4}{a^2 y^3}$$

129. Given $y = \sin^{-1} x$

$$\Rightarrow \frac{dy}{dx} = \frac{1}{\sqrt{1-x^2}}$$

$$\Rightarrow \frac{d^2y}{dx^2} = -\frac{(-2x)}{2(1-x^2)\sqrt{1-x^2}}$$

$$\Rightarrow (1-x^2)\frac{d^2y}{dx^2} = \frac{x}{\sqrt{1-x^2}} = x\frac{dy}{dx}$$

$$\Rightarrow (1-x^2)\frac{d^2y}{dx^2} - x\frac{dy}{dx} = 0$$

$$\Rightarrow (1-x^2)y_2 - xy_1 = 0$$

Hence, the result.

130. **

131. **

132. Given $x = a(\theta + \sin \theta)$, $y = a(1 + \cos \theta)$

$$\Rightarrow \frac{dy}{dx} = \frac{-a \sin \theta}{a(1 + \cos \theta)} = -\frac{\sin \theta}{1 + \cos \theta}$$

$$\Rightarrow \frac{dy}{dx} = -\frac{2\sin\left(\frac{\theta}{2}\right)\cos\left(\frac{\theta}{2}\right)}{2\cos^2\left(\frac{\theta}{2}\right)} = -\tan\left(\frac{\theta}{2}\right)$$

$$\Rightarrow \frac{d^2y}{dx^2} = -\frac{1}{2}\sec^2\left(\frac{\theta}{2}\right) \times \frac{1}{a(1 + \cos \theta)}$$

$$\Rightarrow \frac{d^2y}{dx^2} = -\frac{1}{a} \frac{1}{(1 + \cos \theta)^2}$$

$$\Rightarrow \frac{d^2y}{dx^2} = -\frac{1}{a} \times \frac{a^2}{y^2} = -\frac{a}{y^2}$$

Hence, the result.

133. Given $x = a(1 - \cos \theta)$, $y = a(\theta + \sin \theta)$,

$$\frac{dy}{dx} = \frac{a(1 + \cos \theta)}{a \sin \theta}$$

$$\frac{dy}{dx} = -\frac{2\cos^2\left(\frac{\theta}{2}\right)}{2\sin\left(\frac{\theta}{2}\right)\cos\left(\frac{\theta}{2}\right)} = \cot\left(\frac{\theta}{2}\right)$$

$$\frac{d^2y}{dx^2} = \frac{d}{d\theta}\left(\cot\left(\frac{\theta}{2}\right)\right) \times \frac{d\theta}{dx}$$

$$\frac{d^2y}{dx^2} = -\frac{1}{2}\operatorname{cosec}^2\left(\frac{\theta}{2}\right) \times \frac{1}{a \sin \theta}$$

$$\frac{d^2y}{dx^2} = -\frac{1}{2a} \frac{\operatorname{cosec}^2\left(\frac{\theta}{2}\right)}{\sin \theta}$$

$$\left(\frac{d^2y}{dx^2}\right)_{\theta=\frac{\pi}{2}} = -\frac{1}{2a} \times \frac{2}{1} = -\frac{1}{a}$$

134. Given $y = x \log\left(\frac{x}{a+bx}\right)$

$$\Rightarrow \frac{y}{x} = \log\left(\frac{x}{a+bx}\right)$$

$$\Rightarrow \frac{y}{x} = \log x - \log(a+bx)$$

$$\Rightarrow \frac{x \cdot \frac{dy}{dx} - y}{x^2} = \frac{1}{x} - \frac{b}{a+bx} = \frac{a+bx-bx}{x(a+bx)}$$

$$\Rightarrow x \frac{dy}{dx} - y = \frac{ax^2}{x(a+bx)} = \frac{ax}{a+bx}$$

$$\Rightarrow x \frac{d^2y}{dx^2} + \frac{dy}{dx} \cdot 1 - \frac{dy}{dx} = \frac{(a+bx)a \cdot -ax \cdot b}{(a+bx)^2}$$

$$\Rightarrow x \frac{d^2y}{dx^2} = \frac{a^2}{(a+bx)^2}$$

$$\Rightarrow x^3 \frac{d^2y}{dx^2} = \frac{a^2 x^2}{(a+bx)^2} = \left(\frac{ax}{a+bx}\right)^2$$

$$\Rightarrow x^3 \frac{d^2y}{dx^2} = \left(x \frac{dy}{dx} - y\right)^2$$

135. Given $\sqrt{x+y} + \sqrt{y-x} = c$

$$\Rightarrow \frac{1}{2\sqrt{x+y}}\left(1 + \frac{dy}{dx}\right) + \frac{1}{2\sqrt{y-x}}\left(\frac{dy}{dx} - 1\right) = 0$$

$$\Rightarrow \frac{dy}{dx} = \frac{(\sqrt{x+y} - \sqrt{y-x})(\sqrt{x+y} + \sqrt{y-x})}{c(\sqrt{x+y} + \sqrt{y-x})}$$

$$\Rightarrow \frac{dy}{dx} = \frac{(x+y-y+x)}{c^2}$$

$$\Rightarrow \frac{dy}{dx} = \frac{2x}{c^2}$$

$$\Rightarrow \frac{d^2y}{dx^2} = \frac{2}{c^2}$$

136. We have $\frac{\cos^4 \alpha}{x} + \frac{\sin^4 \alpha}{y} = \frac{1}{x+y}$

$$\Rightarrow \left(\frac{x+y}{x}\right)\cos^4 \alpha + \left(\frac{x+y}{y}\right)\sin^4 \alpha = 1$$

$$\Rightarrow \left(1 + \frac{y}{x}\right)\cos^4 \alpha + \left(1 + \frac{x}{y}\right)\sin^4 \alpha = 1$$

$$\Rightarrow \left(\frac{y}{x}\right)\cos^4 \alpha + \left(\frac{x}{y}\right)\sin^4 \alpha + (\sin^4 \alpha + \cos^4 \alpha) = 1$$

$$\Rightarrow \left(\frac{y}{x}\right)\cos^4 \alpha + \left(\frac{x}{y}\right)\sin^4 \alpha + 1 - 2\sin^2 \alpha \cos^2 \alpha = 1$$

$$\Rightarrow \left(\frac{y}{x}\right)\cos^4 \alpha + \left(\frac{x}{y}\right)\sin^4 \alpha - 2\sin^2 \alpha \cos^2 \alpha = 0$$

$$\Rightarrow \left(\sqrt{\frac{y}{x}}\cos^2 \alpha\right)^2 + \left(\sqrt{\frac{x}{y}}\sin^2 \alpha\right)^2 - 2\left(\sqrt{\frac{x}{y}}\sin^2 \alpha\right)\left(\sqrt{\frac{y}{x}}\cos^2 \alpha\right) = 0$$

$$\Rightarrow \left(\sqrt{\frac{x}{y}} \sin^2 \alpha - \sqrt{\frac{y}{x}} \cos^2 \alpha \right)^2 = 0$$

$$\Rightarrow \left(\sqrt{\frac{x}{y}} \sin^2 \alpha - \sqrt{\frac{y}{x}} \cos^2 \alpha \right) = 0$$

$$\Rightarrow \sqrt{\frac{x}{y}} \sin^2 \alpha = \sqrt{\frac{y}{x}} \cos^2 \alpha$$

$$\Rightarrow y = (x) \tan^2 \alpha$$

$$\Rightarrow \frac{dy}{dx} = \tan^2 \alpha$$

137. Given $y = x \sin x$

$$\Rightarrow \frac{dy}{dx} = x \cos x + \sin x$$

$$\Rightarrow \frac{d^2y}{dx^2} = \cos x - x \sin x + \cos x$$

$$\Rightarrow \frac{d^2y}{dx^2} = 2 \cos x - x \sin x$$

$$\begin{aligned} \text{Now, } x^2 \frac{d^2y}{dx^2} - 2x \frac{dy}{dx} + (x^2 + 2)y &= x^2(2 \cos x - x \sin x) - 2x(x \cos x + \sin x) \\ &\quad + (x^2 + 2)x \sin x \\ &= x^2(2 \cos x - x \sin x + x \sin x - 2 \cos x) \\ &\quad + x(2 \sin x - 2 \sin x) \\ &= x^2(0) + x(0) = 0 \end{aligned}$$

Hence, the result.

138. Given $y = f(x) = x^2$

$$\Rightarrow \frac{dy}{dx} = 2x$$

$$\Rightarrow \frac{d^2y}{dx^2} = 2$$

Also, $x^2 = y$

$$\Rightarrow 2x \frac{dx}{dy} = 1$$

$$\Rightarrow 2x \frac{d^2x}{dy^2} + 2 \left(\frac{dx}{dy} \right)^2 = 0$$

$$\Rightarrow x \frac{d^2x}{dy^2} + \left(\frac{dx}{dy} \right)^2 = 0$$

$$\Rightarrow x \frac{d^2x}{dy^2} + \frac{1}{4x^2} = 0$$

$$\Rightarrow \frac{d^2x}{dy^2} = -\frac{1}{4x^3}$$

$$\text{Now, } \left(\frac{d^2y}{dx^2} \right) \times \left(\frac{d^2x}{dy^2} \right) = 2 \times -\frac{1}{4x^3} = -\frac{1}{2x^3}$$

139. Given $P(x) = ax^3 + bx^2 + cx + d$

$$P'(x) = 2ax^2 + 2bx + c$$

$$P''(x) = 4ax + 2b$$

$$P''(0) = 10 \Rightarrow 2b = 10 \Rightarrow b = 5$$

$$P(0) = -2 \Rightarrow d = -2$$

$$P(1) = -2 \Rightarrow a + b + c + d = -2$$

$$a + 5 + c - 2 = -2$$

$$a + c = -5$$

$$P'(0) = -1 \Rightarrow c = -1$$

From (i), we get, $a = -4$

Hence, the value of $(a + b + c + d + 10)$

$$= -4 + 5 - 1 - 2 + 10$$

$$= 8$$

140. Given $f(x) = x + \tan x$

Since f is the inverse of g , so $(f \circ g)(x) = x$

$$f(g(x)) = x$$

$$f'(g(x))g'(x) = 1$$

$$g'(x) = \frac{1}{f'(g(x))}$$

$$g'(x) = \frac{1}{1 + \sec^2(g(x))} = \frac{1}{1 + 1 + \tan^2(g(x))}$$

$$g'(x) = \frac{1}{2 + \tan^2(g(x))}$$

141. Given $f(x) = e^x + x^3 - 1$

$$f'(x) = e^x + 3x^2$$

$$f''(x) = e^x + 6x$$

when $y = f(x) = e$, then $x = 1$

$$g''(x) = -\frac{f''(x)}{\{f'(x)\}^3}$$

$$g''(e) = -\frac{f''(e)}{\{f'(e)\}^3} = -\frac{e + 6}{(e + 3)^3}$$

142. Given $f(x) = 1 + x^3$

$$\Rightarrow f'(x) = 3x^2$$

$$\Rightarrow f''(x) = 6x$$

$$\Rightarrow f'''(x) = 6$$

when $y = 2$, then $x = 1$

$$\text{Now, } g''(y) = -\frac{f''(x)}{\{f'(x)\}^2}$$

$$\Rightarrow g'''(y) = -\frac{f'(x)f'''(x) - 3\{f''(x)\}^2}{\{f'(x)\}^3}$$

$$\Rightarrow g'''(2) = -\frac{f'(1)f'''(1) - 3\{f''(1)\}^2}{\{f'(1)\}^3}$$

$$\Rightarrow g'''(2) = -\frac{3 \times 6 - 3(6)^2}{(3)^3}$$

$$\Rightarrow g'''(2) = \frac{72}{27} = \frac{8}{3}$$

143. Given $f(g(x)) = x$

$$\Rightarrow f'(g(x))g'(x) = 1$$

$$\Rightarrow g'(x) = \frac{1}{f'(g(x))}$$

$$\Rightarrow g'(x) = \frac{1}{1 + (f(g(x)))^2}$$

$$\Rightarrow g'(x) = \frac{1}{1 + x^2}$$

Hence, the result.

144 We have $f(x) = \begin{vmatrix} x^3 & \sin x & \cos x \\ 6 & -1 & 0 \\ p & p^2 & p^3 \end{vmatrix}$

$$\Rightarrow f'(x) = \begin{vmatrix} 3x^2 & \cos x & -\sin x \\ 6 & -1 & 0 \\ p & p^2 & p^3 \end{vmatrix}$$

$$\Rightarrow f''(x) = \begin{vmatrix} 6x & -\sin x & -\cos x \\ 6 & -1 & 0 \\ p & p^2 & p^3 \end{vmatrix}$$

$$\Rightarrow f'''(x) = \begin{vmatrix} 6 & -\cos x & \sin x \\ 6 & -1 & 0 \\ p & p^2 & p^3 \end{vmatrix}$$

$$\Rightarrow f'''(x) = \begin{vmatrix} 6 & -1 & 0 \\ 6 & -1 & 0 \\ p & p^2 & p^3 \end{vmatrix} = 0$$

Thus, $\frac{d^3}{dx^3}(0) = 0$

145. Let $f(x) = \begin{vmatrix} (1+x)^{a_1b_1} & (1+x)^{a_1b_2} & (1+x)^{a_1b_3} \\ (1+x)^{a_2b_1} & (1+x)^{a_2b_2} & (1+x)^{a_2b_3} \\ (1+x)^{a_3b_1} & (1+x)^{a_3b_2} & (1+x)^{a_3b_3} \end{vmatrix}$

Differentiating both sides w.r.t x and then putting $x = 0$, we get, the co-efficient of x

$$\begin{vmatrix} a_1b_1 & a_1b_2 & a_1b_3 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{vmatrix} + \begin{vmatrix} 1 & 1 & 1 \\ a_2b_1 & a_2b_2 & a_2b_3 \\ 1 & 1 & 1 \end{vmatrix}$$

$$+ \begin{vmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ a_3b_1 & a_3b_2 & a_3b_3 \end{vmatrix}$$

$$= 0 + 0 + 0$$

$$= 0.$$

146. We have $f(x) = \begin{vmatrix} x+1 & x+2 & x+3 \\ x+4 & x+5 & x+6 \\ x+7 & x+8 & x+9 \end{vmatrix}$

$$\Rightarrow f'(x) = \begin{vmatrix} 1 & 1 & 1 \\ x+4 & x+5 & x+6 \\ x+7 & x+8 & x+9 \end{vmatrix}$$

$$+ \begin{vmatrix} x+1 & x+2 & x+3 \\ 1 & 1 & 1 \\ x+7 & x+8 & x+9 \end{vmatrix}$$

$$+ \begin{vmatrix} x+1 & x+2 & x+3 \\ x+4 & x+5 & x+6 \\ 1 & 1 & 1 \end{vmatrix}$$

$$\Rightarrow f''(x) = 0$$

On integration, we get,

$$\Rightarrow f(x) = c_1$$

Again, integrating, we get,

$$\Rightarrow f(x) = c_1x + c_2$$

Hence, the result.

147. We have $\begin{vmatrix} 1 & 2x & 3x^2 \\ x & x^2 & x^3 \\ 0 & 2 & 6x \end{vmatrix}$

$$\Rightarrow f'(x) = \begin{vmatrix} 0 & 2 & 6x \\ x & x^2 & x^3 \\ 0 & 2 & 6x \end{vmatrix} + \begin{vmatrix} 1 & 2x & 3x^2 \\ 1 & 2x & 3x^2 \\ 0 & 2 & 6x \end{vmatrix}$$

$$+ \begin{vmatrix} 1 & 2x & 3x^2 \\ x & x^2 & x^3 \\ 0 & 0 & 6 \end{vmatrix}$$

$$\Rightarrow f'(x) = 0 + 0 + 6(x^2 - 2x^2) = -6x^2$$

$$\Rightarrow f'(1) = -6$$

148. We have

$$f(x) = \begin{vmatrix} \cos x & \sin x & \cos x \\ \cos 2x & \sin 2x & 2\cos 2x \\ \cos 3x & \sin 3x & 2\cos 3x \end{vmatrix}$$

$$\Rightarrow f'(x) = \begin{vmatrix} -\sin x & \cos x & -\sin x \\ \cos 2x & \sin 2x & 2\cos 2x \\ \cos 3x & \sin 3x & 2\cos 3x \end{vmatrix}$$

$$+ \begin{vmatrix} \cos x & \sin x & \cos x \\ -2\sin 2x & 2\cos 2x & -4\sin 2x \\ \cos 3x & \sin 3x & 2\cos 3x \end{vmatrix}$$

$$+ \begin{vmatrix} \cos x & \sin x & \cos x \\ \cos 2x & \sin 2x & 2\cos 2x \\ -3\sin 3x & 3\cos 3x & -6\sin 3x \end{vmatrix}$$

$$\Rightarrow f'\left(\frac{\pi}{2}\right) = \begin{vmatrix} -1 & 0 & -1 \\ -1 & 0 & -2 \\ 0 & -2 & 0 \end{vmatrix} + \begin{vmatrix} 0 & 1 & 0 \\ 0 & -2 & 0 \\ 0 & -1 & 0 \end{vmatrix} + \begin{vmatrix} 0 & 1 & 0 \\ -1 & 0 & -2 \\ 3 & 0 & 6 \end{vmatrix}$$

$$\Rightarrow f'\left(\frac{\pi}{2}\right) = (2 - 1) - 1(6 - 6) = 1$$

149. We have $y = x^n$

$$\Rightarrow y_1 = e^x$$

$$\Rightarrow y_2 = e^x$$

$$\Rightarrow y_3 = e^x$$

$$\dots$$

$$\dots$$

$$\Rightarrow y_n = e^x$$

$$\text{Thus, } y_n(0) = e^0 = 1$$

150. We have $y = a^x$

$$\Rightarrow y_1 = a^x \log a$$

$$\Rightarrow y_2 = a^x (\log a)^2$$

$$\Rightarrow y_3 = a^x (\log a)^3$$

$$\dots$$

$$\dots$$

$$\Rightarrow y_n = a^x (\log a)^n$$

$$\text{Thus, } y_n(0) = a^0 (\log a)^n = (\log a)^n$$

151. We have $y = \frac{1}{(x+1)}$

$$\Rightarrow y_1 = -\frac{1}{(x+1)^2}$$

$$\Rightarrow y_2 = \frac{2}{(x+1)^3}$$

$$\Rightarrow y_3 = -\frac{6}{(x+1)^4} = \frac{(-1)^3 3!}{(x+1)^{3+1}}$$

$$\Rightarrow y_4 = \frac{(-1)^4 4!}{(x+1)^{4+1}}$$

$$\dots$$

$$\dots$$

$$\Rightarrow y_n = \frac{(-1)^n (n)!}{(x+1)^{n+1}}$$

$$\text{Thus, } y_n(0) = \frac{(-1)^n (n)!}{(0+1)^{n+1}} = (-1)^n (n)! \quad \text{152. We have } y = \sin x$$

$$\Rightarrow y_1 = \cos x = \sin\left(\frac{\pi}{2} + x\right)$$

$$\Rightarrow y_2 = -\sin x = \sin\left(2 \cdot \frac{\pi}{2} + x\right)$$

$$\Rightarrow y_3 = -\cos x = \sin\left(3 \cdot \frac{\pi}{2} + x\right)$$

$$\dots$$

$$\dots$$

$$\dots$$

$$\Rightarrow y_n = \sin\left(\frac{n\pi}{2} + x\right)$$

$$\text{Thus, } y_n(0) = \sin\left(\frac{n\pi}{2} + 0\right) = \sin\left(\frac{n\pi}{2}\right)$$

$$= \begin{cases} 1 & : n = (4k+1), k \in \mathbb{I} \\ -1 & : n = (4k-1), k \in \mathbb{I} \\ 0 & : n = 2k, k \in \mathbb{I} \end{cases}$$

153. We have $y = x^n$

$$y_1 = nx^{n-1}$$

$$y_2 = n(n-1)x^{n-2}$$

$$y_3 = n(n-1)(n-2)x^{n-3}$$

$$\dots$$

$$\dots$$

$$y_k = n(n-1)(n-2)\dots(n-(k-1))x^{n-k+1}$$

$$\text{Thus, } y_k(1) = n(n-1)(n-2)\dots(n-(k-1)) = (n-(k-1))!$$

154. We have $y = A \sin x + B \cos x$

$$\Rightarrow y_1 = A \cos x - B \sin x$$

$$\Rightarrow y_2 = -A \sin x - B \cos x = -y$$

$$\Rightarrow y_2 + y = 0$$

By Leibnitz theorem, we get,

$$\Rightarrow y_{n+2} + (0)y_{n+1} + (0)y_n + y_n = 0$$

$$\Rightarrow y_{n+2} + y_n = 0$$

155. We have $y = \tan^{-1}x$

$$\Rightarrow y_1 = \frac{1}{(1+x^2)}$$

$$\Rightarrow (1+x^2)y_1 = 1$$

$$\Rightarrow (1+x^2)y_2 + 2xy_1 = 0$$

By Leibnitz theorem, we get,

$$(1+x^2)y_{n+2} + (2(n+1)x)y_{n+1} + \left(\frac{2n(n-1)}{2}\right)y_n + 2xy_{n+1} + (2n)y_n = 0$$

$$\Rightarrow (1+x^2)y_{n+2} + (2(n+2)x)y_{n+1} + n(n+1)y_n = 0$$

156 We have $y = \sin^{-1}x$

$$\Rightarrow y_1 = \frac{1}{\sqrt{1-x^2}}$$

$$\Rightarrow y_2 = -\frac{-2x}{2(1-x^2)\sqrt{1-x^2}} = \frac{xy_1}{(1-x^2)}$$

$$\Rightarrow (1-x^2)y_2 = xy_1$$

$$\Rightarrow (1-x^2)y_2 - xy_1 = 0$$

By Leibnitz theorem, we get,

$$(1-x^2)y_{n+2} - 2(n+1)xy_{n+1} - n(n-1)y_n - xy_{n+1} - ny_n = 0$$

$$\Rightarrow (1-x^2)y_{n+2} - (2n+1)xy_{n+1} - n^2y_n = 0$$

157. We have $y = a \cos(\log x) + b \sin(\log x)$

$$\Rightarrow y_1 = -a \sin(\log x) \cdot \frac{1}{x} + b \cos(\log x) \cdot \frac{1}{x}$$

$$\Rightarrow xy_1 = -a \sin(\log x) + b \cos(\log x)$$

$$\Rightarrow xy_2 + y_1 = -a \cos(\log x) \cdot \frac{1}{x} - b \sin(\log x) \cdot \frac{1}{x}$$

$$\Rightarrow x^2y_2 + xy_1 = -y$$

$$\Rightarrow x^2y_2 + xy_1 + y = 0$$

By Leibnitz theorem, we get,

$$\begin{aligned} & x^2y_{n+2} + 2(n+1)xy_{n+1} + n(n-1)y_n \\ & + xy_{n+1} + ny_n + y_n = 0 \\ \Rightarrow & x^2y_{n+2} + (2n+3)xy_{n+1} + (n^2+1)y_n = 0 \end{aligned}$$